

How important are climate change risks for predicting clean energy stock prices? Evidence from machine learning predictive modeling and interpretation

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Abstract

The clean energy equity sector plays an important role in the transition to a low-carbon economy. This paper explores the role of climate change risks in predicting the direction of clean energy stock prices (solar, wind, nuclear). We employ machine learning models, including random forests, boosting, extremely randomized trees, and support vector machines, to make our predictions. Variable importance is determined using Shapley/SHAP values. Notably, tree-based ensemble and boosting models show an accuracy exceeding 85% for the 10-day to 20-day forecast period. For the stock prices of solar, wind, and nuclear energy, inflation expectations and technical indicators (which account for behavioral factors) such as on-balance volume and Williams' accumulation/distribution are important features within this forecast range. For wind and solar energy stocks moving averages are also important additional features while for nuclear energy stocks economic policy uncertainty and stock market volatility are additional important features. In the five day to twenty day forecast horizon, climate change risks are not important features. These results align with a body of literature that raises concerns about equity prices not fully reflecting climate change risks. An equally weighted portfolio of wind, solar, and nuclear energy stock prices that used trading signals from an Extra Trees prediction model outperformed a buy and hold portfolio in terms of risk adjusted returns. These results are robust to trading costs and weekly or monthly portfolio rebalancing.

Keywords: forecasting; random forests; clean energy stock prices; machine learning; climate change risks; sustainable finance

JEL Classification: G17; Q42, Q47, Q54

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1. Introduction

According to NASA, the year 2023 was the hottest on record¹. Global temperatures were 1.2 degrees Celsius above the average for the baseline period of 1951 to 1980. Increases in temperature are primarily being driven by the burning of fossil fuels (Field et al., 2012). Climate change induced by temperature increases leads to extreme weather events like heat waves, droughts, flooding, rising sea levels, increased wind activity, and forest fires². Extreme weather events can affect physical capital, supply chains, financial stability and overall economic activity. Extreme weather events can be very costly. In the United States, climate disasters cost \$95 billion in damages in 2020³. Since 1980, climate disasters in the United States have cost more than \$2 trillion. The magnitude of future risks from climate change are going to depend upon how society responds. For example, in looking at flood losses for the world's 136 largest coastal cities Hallegatte et al. (2013) find that the global average annual losses would exceed \$1 trillion in 2050 in a scenario without investment in climate change adaptation and \$60 billion in a scenario with adaptation. The WEF (2024) in looking at global risks ranks extreme weather events number two in severity over the next two years and number one over the next ten years. The data for this ranking comes from the WEF's Global Risks Perception Survey which collects insights from 1490 experts in academics, business, government, and international organizations. In response to climate change risks, there is a developing literature that looks at the impact of physical and transition risks related to climate change on stock prices (Ardia et al., 2023; Bolton and Kacperczyk, 2023, 2021; Campiglio et al., 2022; Eren et al., 2022; Reboredo and Ugolini, 2022; Venturini, 2022). Physical risks arise from the direct physical costs that climate change

¹ <https://www.nasa.gov/news-release/nasa-analysis-confirms-2023-as-warmest-year-on-record/>

² <https://www.nrdc.org/stories/what-are-effects-climate-change#weather>

³ <https://www.brookings.edu/articles/what-is-climate-risk-and-why-does-it-matter/>

imposes on business and society. Transition risks are associated with climate change mitigation and adoption policy, emerging new clean energy technologies, and behavioral changes of consumers and investors.

The clean energy equity sector plays an important role in the transition to a low carbon economy and forecasting clean energy stock prices are important for constructing and risk management of investment portfolios, raising capital for new or existing clean energy companies, and measuring the general outlook for the clean energy sector. While there is some research on forecasting clean energy stock prices (Ghosh and Jana, 2024; Sadorsky, 2022a, 2022b, 2021), none of these papers study how important climate change risks are for predicting clean energy stock prices.

According to asset pricing theory, stock prices are equal to the present value of expected future cash flows (Cochrane, 2005). In addition investors need compensation for any risk they take on (Pástor and Veronesi, 2013). Changes in stock prices will reflect changes in expected future cash flows and changes in risk exposure. If climate change risks affect future expected cash flows and risk then climate change risks will affect stock prices (Campiglio et al., 2022). There are several ways in which climate change transition risks can affect clean energy stock prices. First, higher climate change risks are likely to accelerate the transition to a low carbon economy and this increases the cash flow to clean energy stocks because these companies are directly involved in creating the transition (a greater usage of renewable energy in the fuel mix). Second, clean energy stocks have less CO₂ emissions and are less exposed to carbon pricing risk and other regulations specifically aimed at reducing CO₂ emissions. Clean energy stocks are thus less risky compared to high CO₂ emitters (Alessi et al., 2021; Bolton and Kacperczyk, 2023, 2021). Higher cash flows and lower risk results in higher prices for clean energy stocks. Third,

green stocks can outperform brown stocks when concerns about climate change strengthen unexpectedly shifting investor preferences towards green firms and away from brown firms (Pástor et al., 2021). Transition risk has been found to have a positive impact on clean energy stock prices (Ardia et al., 2023; Campiglio et al., 2022; Reboredo and Ugolini, 2022). The impact of physical risks on clean energy companies and their stock prices is expected to be positive. Large scale physical risks arising from extreme weather events provides a powerful incentive to address climate change by increasing the usage of clean energy which increases the cash flows of these companies.

If climate change risks are important features for predicting clean energy stock prices, then forecasting models that omit climate change risks will produce forecasts that lead to sub-optimal decision making. While evidence of predictability is inconsistent with the efficient markets hypothesis (Malkiel, 2003), predictability can be explained by momentum and behavior biases (Christoffersen and Diebold, 2006; Gray and Vogel, 2016; Lo et al., 2000; Moskowitz et al., 2012) and the adaptive market hypothesis (Lo, 2017). However, it may be the case that climate change risks are not important features when forecasting clean energy stock prices and the omission of climate change risk features will have little impact on forecast accuracy. This could happen if equity prices are not fully reflecting climate change risks (Bolton et al., 2020; Campiglio et al., 2022; Eren et al., 2022; IMF, 2020).

The purpose of this article is to investigate the importance of climate change risks for predicting clean energy stock prices. Exchange traded funds (ETFs) representing solar energy, wind energy, and nuclear energy are used in this research. ETFs are low cost and widely traded financial assets that are used by individual and institutional investors to construct portfolios. Solar and wind ETFs are used in the analysis because solar and wind are the two largest

components of the renewable energy sector and a nuclear ETF is included because there is a renewed interest in nuclear energy as government attitudes towards nuclear energy have recently become more favorable. Analysis is conducted using the direction of clean energy stock prices rather than actual prices because in practice, investors are more interested in the direction of asset prices rather than the actual values when determining asset allocation (Pesaran and Timmermann, 2002). Numerous studies have reported high accuracy from predicting asset price direction in general (Ballings et al., 2015; Basak et al., 2019; Leung et al., 2000; Lohrmann and Luukka, 2019; Nyberg, 2011; Nyberg and Pönkä, 2016; Pönkä, 2016) and clean energy stock prices in particular (Sadorsky, 2022a, 2022b, 2021). Machine learning models are employed to predict the price direction of clean energy stocks because compared to standard regression models, machine learning models can achieve higher prediction accuracy when the relationships between the predictor and features are complex (Hastie et al., 2009; James et al., 2022; Mullainathan and Spiess, 2017). Machine learning techniques like support vector machines and random forests have demonstrated high accuracy when predicting stock prices (Ampomah et al., 2020; Ballings et al., 2015; Basak et al., 2019; Ghosh et al., 2021; Khan et al., 2020; Lohrmann and Luukka, 2019; Weng et al., 2018). Building on this existing literature, the machine learning techniques used in this present paper include random forests, extremely randomized trees, LightGBM, support vector machine, Lasso, and Naïve Bayes. The features (also referred to as explanatory variables or predictors) include technical indicators (to capture behavioral factors), macroeconomic variables (interest rates, inflation expectations, market volatility – to account for business cycle conditions), and climate change risks. The relevance of technical indicators and macroeconomic variables for predicting clean energy stock prices (or price direction) has been described and established (Ghosh and Jana, 2024; Sadorsky, 2022b, 2022b, 2022a). Forecasts

are produced for a range of days (one day to twenty days) to show how prediction accuracy varies with the forecast horizon. A twenty day forecast horizon corresponds approximately to the number of trading days in a calendar month. Rather than relying on one model and one forecast horizon, multiple models are estimated, and forecasts produced from one day to twenty days to see how robust the results are to model choice and forecast horizon. A portfolio comparison was used to demonstrate the practical implications of the forecasts. More specifically, an Extra Trees model was used to generate trading signals and a portfolio that used these signals was compared to a buy and hold portfolio.

Several important findings emerge from the analysis reported in this paper. First, with respect to forecast accuracy, tree-based ensemble and boosting models have the highest accuracy. The accuracy of tree-based ensemble and boosting models ranges between 85% and 90% for the 10-day to 20-day forecast period. Second, in the five day to twenty day forecast horizon, climate change risks are not an important predictor of any of the clean energy assets studied. These results are consistent with a body of literature that raises concerns about equity prices not fully reflecting climate change risks (Bolton et al., 2020; Campiglio et al., 2022; Eren et al., 2022; IMF, 2020). Third, for ETFs representing solar, wind, and nuclear energy, inflation expectations and technical indicators (which account for behavioral factors) such as on-balance volume and Williams' accumulation/distribution are important features within this forecast range. For wind and solar stocks, moving averages are additional important features while for the nuclear energy ETF economic policy uncertainty and the VIX volatility indicator are also important features. Fourth, an equally weighted portfolio of wind, solar, and nuclear ETFs that used trading signals from an Extra Trees prediction model greatly outperformed a buy and hold

portfolio in terms of risk adjusted returns. These results are robust to reasonable trading costs and weekly or monthly portfolio rebalancing.

The remainder of this paper is organized as follows. The next section of the paper presents some background literature. The machine learning models are described in Section 3. The data are presented in Section 4. Section 5 reports the results, a discussion of the findings is provided in Section 6, while Section 7 concludes the paper.

2. Background literature

There is a developing literature that looks at the impact of physical and transition risks related to climate change on stock prices (Campiglio et al., 2022; Eren et al., 2022; Venturini, 2022). Physical risks arise from the direct physical costs that climate change imposes on business and society. Transition risks are associated with climate change mitigation and adoption policy, emerging new clean energy technologies, and behavioral changes of consumers and investors. Physical risks lead to higher operating costs for firms which affects profitability and stock prices (Campiglio et al., 2022). Some research finds investors do pay attention to climate risk and that a carbon premium exists whereby higher polluting companies earn higher returns. Bolton and Kacperczyk (2021) find that after controlling for size, book-to-market, and other factors US firms with higher carbon dioxide emissions (CO₂) earn higher returns compared to the stocks of firms less exposed to CO₂. Bolton and Kacperczyk (2023) extend their analysis to a sample of global companies (14,400 firms in 77 countries) and find higher stock returns associated with higher levels and growth rates of CO₂ in all sectors and most countries. Carbon premia are higher in countries with stricter domestic climate policies.

There is evidence that stocks with less carbon risk outperform stocks with higher carbon risk. Reboredo and Ugolini (2022) find that stocks with lower transition risk outperform stocks

with higher transition risk. In their survey of the literature, Campiglio et al. (2022) find that climate change risk mostly has a negative impact on stock prices but in the case of renewable energy firms, transition risk appears to be priced as transition risk has a positive impact on stock prices. Ardia et al. (2023) test the model of Pastor et al. (2021) and find that green stocks outperform brown stocks when concerns about climate change increases unexpectedly. In studying the impact of climate change risks on stock prices, Faccini et al. (2023), building on the work of Engle et al. (2020), consider novel risk measures related to; natural disasters, global warming, international summits, and U.S. climate policy, respectively. They find that only the climate-policy factor is priced, especially post-2012. Basher and Sadorsky (2024) find that climate change risks are not an important determinant of systemic risk in a network of stocks that includes clean energy, electric vehicles, and critical minerals. Bouri et al. (2023) examine the impact of climate risks on the nexus of clean energy and technology stocks using a time-varying correlation model. Physical and transition climate risks are positively associated with the long-term correlation between clean energy and technology stock indices, whereas the effect of transition risk is more robust to different sample periods and alternative stock indices. Pham et al. (2023) find that green stocks respond positively to favorable climate change policy like the Paris Agreement or Biden becoming President of the United States. Green firms in light industries with high litigation risk, small green firms, and green firms with high idiosyncratic volatility tend to exhibit stronger cumulative reactions when compared with other green firms. In studying stocks across the G7, Bauer et al. (2022) find that green stocks have higher returns than brown stocks. Antoniuk and Leirvik (2024) find that events related to climate change can impact stock prices. Clean energy stocks benefited from the Paris Agreement, Climategate, and the Fukushima disaster. Fossil fuel stocks benefited from weaker climate change policy.

The research that comes closest to this present study are the studies by Sadorsky (Sadorsky, 2022a, 2022b, 2021) and Ghosh and Jana (2024). Sadorsky (2021) uses a random forests model to predict the price direction of three clean energy ETFs (PBW, ICLN, QCLN). He finds that tree bagging and random forests have higher predictive accuracy than logit models. The feature space includes technical indicators but does not include business cycle variables. Sadorsky (2022a) uses support vector machine and random forests to predict solar stock prices. Random forests, extremely randomized trees, tree bagging, and support vector machine produce more accurate predictions than logit or boosted logit. The feature space includes technical indicators and the volatility of oil prices and silver prices. The most important predictors are technical indicators. Non-technical features like the volatility of oil prices and the volatility of silver prices rank in the top third. Sadorsky (2022b) builds on Sadorsky (2021) by using a wider selection of machine learning models and includes features for market volatility and economic policy uncertainty. Tree-based models have high accuracy in the ten-day to twenty-day forecast range. Technical indicators are overall, the most important features. Market volatility variables are also important features, but economic policy uncertainty has little importance. In related research Ghosh and Jana (2024) use Facebook's Prophet and NeuralProphet to predict the stock prices of seven clean energy sub-sectors. They find that technical indicators are the most important features which is consistent with the findings of Sadorsky (2022a, 2022b, 2021). None of these papers, however, study how important climate change risks are for predicting clean energy stock prices.

3. Methods

3.1. Machine learning models

This section provides a brief non-technical description of the machine learning models used (random forests, boosting, extremely randomized trees, LightGBM, SVM, Lasso, and Naïve Bayes) to predict clean energy stock price direction. A more technical treatment of these models can be found in Hastie et al. (2009) while a good introduction for practitioners is provided by James et al. (2022). All of the models predict price direction. Price direction is categorized as either “positive” or “non-positive,” where “positive” indicates a positive change in price from one period to the next, and “non-positive” represents a non-positive or zero change. Forecasts are made from one day to twenty days (approximately four weeks of trading data) in order to see how forecast accuracy varies with the forecast horizon. In order to avoid look forward bias, forecasts made at time period t use information available at time period $t-1$. Since this is a classification problem, the benchmark models include the Naïve Bayes classifier and Lasso.

Random forests are ensembles of decision trees (Breiman, 2001). Decision trees can be trained to have high prediction accuracy. However, decision trees can be very sensitive to the data and small changes in the data can yield different prediction results. This is known as high variance. One approach to deal with this problem of high variance is to create ensembles or a forest of decision trees. In practice, however, there is only one data set. Copies of data are made using bootstrap replication and a decision tree fit to each bootstrap data set. Predictions from each decision tree are averaged to yield the overall prediction.

Extremely randomized trees (Extra Trees) are related to random forests but there are some important differences (Geurts et al., 2006). In random forests a decision tree is constructed using a subset of predictors chosen at random and then splits for each of these randomly chosen predictors are compared to determine the best split. The process is repeated until a stopping criteria is met and a tree is complete. The same process is used to construct many trees. In

extremely randomized trees a subset of predictors is chosen at random but in addition a small number of splits (typically one) for each feature is chosen at random. In other words, a subset of predictors is chosen at random and for each predictor a small set of splits is chosen. From this combination the best split is determined. Unlike random forests which grow trees on bootstrap copies of the original data set, extremely randomized trees grow trees using the original data set.

Boosting is another ensemble technique that is used to deal with the high variance problem associated with decision trees (James et al., 2022). Boosting uses gradient optimization to train a sequence of trees. Whereas random forests and extremely randomized trees grow trees in parallel to create a forest, boosting creates a sequential set of trees. In boosting models are created in sequence. The first tree fits the target variable. The second tree fits the residuals of the first tree. The third tree fits the residuals of the second tree, etc. With each new iteration a new model is estimated. The analysis in this paper uses LightGBM (Ke et al., 2017). LightGBM is based on extreme gradient boosting (XGBoost) (Chen and Guestrin, 2016) but is faster. XGBoost improves on gradient boosting by including regularization and stochastic sub-sampling to the boosting process.

Support vector machines (SVMs) separate the two classes of stock price direction by finding the optimal hyperplane that maximizes the margin between the closest data points of opposite classes (Hastie et al., 2009; James et al., 2022). Since multiple hyperplanes can be found to differentiate classes, maximizing the margin between points enables the algorithm to find the best decision boundary between classes. The lines that are adjacent to the optimal hyperplane are known as support vectors as these vectors run through the data points that determine the maximal margin. When the data are not linearly separable, kernel functions are used to transform the data into higher-dimensional space in order to perform linear separation.

3.2. Parameter tuning

Random forests and Extra Trees were estimated with 500 trees and the number of predictors at each cut determined by the square root of the total number of features (James et al., 2022). Random forests are not affected by the number of trees provided a large enough number of trees is chosen. A large number of trees does not cause overfitting, but a small number of trees leads to a high error rate in testing. Preliminary analysis showed that 500 trees was sufficient to capture any large movements in the test error. In the case of extremely randomized trees, 500 trees were estimated, and one random split was selected for each randomly selected feature.

For LightGBM the most important tuning parameters are the number of leaves, the number of iterations, and the learning rate. Based on 10-fold cross-validation with 5 repeats for 10-day forecasts, the number of leaves was set at 40, the maximum number of iterations set at 500, and the learning rate was set at 0.1. The forecasts from LightGBM are robust to small changes (10%) in these parameters.

For SVM a radial basis SVM was fit to the data. The cost and gamma parameters were determined using cross-validation and grid search. A 10-fold cross validation with 10 repeats was used to select the best values for these parameters. The cost parameter was selected from a range of values (0.1, 1, 10, 100, 1000), and gamma was chosen from a range of values (0.2, 0.5, 1, 2, 3, 4).

The Naïve Bayes classifier and Lasso are included in the analysis as benchmarks. The Naïve Bayes classifier with one tuning parameter (kernel) was trained with 5-fold cross-validation. Lasso has one tuning parameter, λ , selected using the 10-fold cross-validation approach described in Friedman et al. (2010).

3.3. Prediction evaluations

The predicted and actual values of price direction are compared using a confusion matrix. In particular, accuracy, Kappa, and F1 are used to compare forecast accuracy. Accuracy is the total number of correct predictions divided by the total number of predictions and ranges from 0 (no predictions are correct) to unity (all predictions are correct). Kappa is related to accuracy and takes into account both the number of agreements and the number of disagreements. Kappa is useful for measuring the consistency of the classification. The F1 score combines precision and recall. Both Kappa and F1 take into account class balance. For evenly balanced classification, all three measures produce similar numeric values.

3.4. Feature importance

Feature importance is estimated using Shapley values. Shapley values are based on concepts used in conditional game theory. The Shapley value represents the contribution of each feature to the predicted value compared to the average prediction for the entire data set (Lundberg and Lee, 2016, 2017). The computation time increases exponentially with the number of features and in many applications exact Shapley values are not possible. In practice the number of feature contributions can be set using simulation. For the analysis reported in this paper 200 simulations were used. A larger number of simulations (500) provided no major differences in the rankings of feature importance.

The workflow is as follows. First, the data were randomly split into a training data set (70% of the data) and testing data set (30% of the data)⁴. Then for each machine learning model

⁴ Bergmeir et al. (2018) show that in the context of autoregressive models, cross validation is valid provided the autocorrelation is not too high. This is not likely to be a problem in data sets like the ones used in this paper where the features contain lagged values and the dependent variable is a classifier. Sadorsky (2022b) uses random forests to predict price direction for clean energy ETFs. He finds that prediction accuracy computed from cross validation is similar to that from time series cross validation.

forecasts are made from one day to twenty days⁵. The forecast accuracy of different models is compared using statistics constructed from the confusion matrix. Shapley values are used to compute feature importance. Figure 1 shows a flowchart of the machine learning process.

⁵ The estimation of Lasso and SVM used scaled data. Naïve Bayes and decision tree models are insensitive to feature scaling.

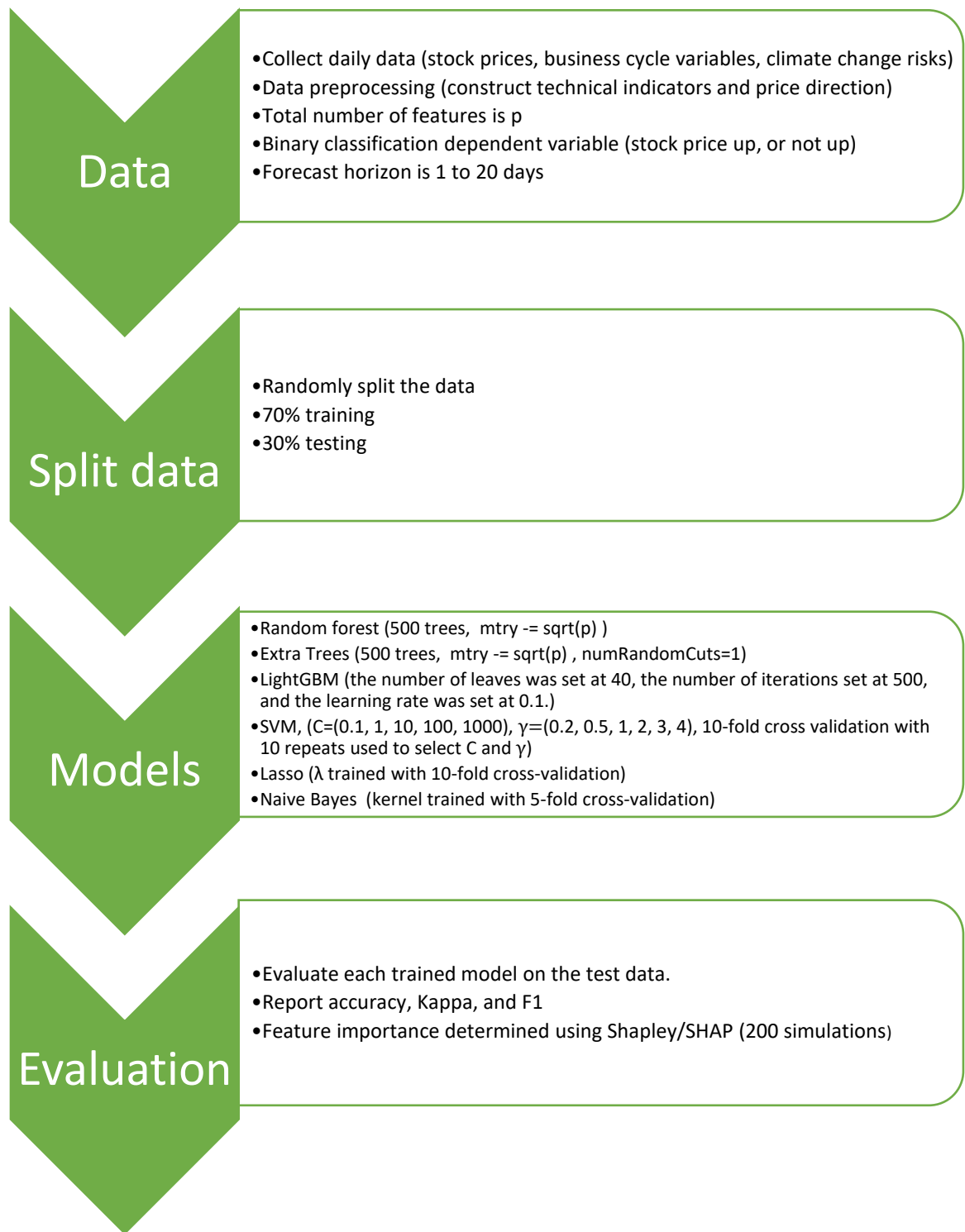


Figure 1. Flowchart of the machine learning process.

All calculations were done using R (R Core Team, 2022). The ranger machine learning package (Wright et al., 2022), the e1071 package (Meyer et al., 2021), the LightGBM package (Shi et al., 2023), the caret package (Kuhn et al., 2020), the quantmod package (Ryan et al., 2023), the fastshap package (Greenwell, 2021) and the pROC package (Robin et al., 2023) were used in the analysis.

4. Data

This study uses data on clean energy stock prices, macroeconomic variables (interest rates, inflation expectations, market volatility, economic policy uncertainty), and climate change risk measures. Clean energy stock prices are measured by three ETFs. FAN is the ticker symbol for the First Trust Global Wind Energy ETF. This ETF holds stocks of companies that specialize in wind energy. TAN is the ticker symbol for the Invesco Solar ETF. This ETF is comprised of stocks of companies that specialize in solar energy. NLR is the ticker symbol for the VanEck Uranium+Nuclear Energy ETF. NLR is a portfolio of stocks of companies involved in uranium and nuclear energy.

Stock prices are equal to the discounted values of expected future cash flows (Cochrane, 2005; Jones and Kaul, 1996). Thus, stock prices are affected by movements in expected cash flows and discount rates. Macroeconomic variables that have an important impact on stock prices include interest rates, inflation, oil prices, and market volatility (Chen et al., 1986).

Interest rates and inflation are important variables that characterize the business cycle and can affect stock prices (Chen et al., 1986; Lee, 1992). There is an inverse relationship between stock prices and interest rates (Kaul, 1987). When interest rates rise bonds become more attractive and investors switch money from stocks into bonds. Higher interest rates also increase

the cost of borrowing money for companies which can put pressure on capital expenditures and cash flow stability. Higher interest rates also affect the discount rate used to discount future cash flows. This study uses several measures of interest rates. The yield on the US three-month Treasury bill (ThreemTbill) represents short-term rates while long term interest rates are represented by the yield on the US ten-year bond (Tenyrbond). The interest rate spread (spread) is the difference between the long interest rate and the short interest rate.

The relationship between inflation and stock prices is complicated (Kaul, 1987). Low to moderate inflation generally indicates a healthy economy. The demand for goods and services is steady and prices are predictable. Higher levels of inflation however create uncertainty and volatility as consumer spending slows. central banks respond to higher inflation by rising interest rates which, as the previous paragraph has described, has a negative impact on stock prices. Break even inflation (be_inflation) and five year expected inflation (Fiveyrinfexp) are included in the analysis to account for inflationary effects.

Economic recessions are the single biggest source of stock market volatility and stock market volatility has a negative impact on stock prices (Hamilton and Lin, 1996; Schwert, 1989). Low stock market volatility environments which coincide with periods of economic expansion are generally more conducive to stock price appreciation. There is a body of literature showing oil prices and the prices of technology stocks affect clean energy stock prices (Basher and Sadorsky, 2024; Bondia et al., 2016; Bouri et al., 2019; Dutta, 2017; Dutta et al., 2018; Ferrer et al., 2018; Geng et al., 2021; Gupta, 2017; Henriques and Sadorsky, 2008; Kumar et al., 2012; Le et al., 2021; Maghyereh et al., 2019; Managi and Okimoto, 2013; Nasreen et al., 2020; Pham, 2021, 2019; Reboredo, 2015; Reboredo et al., 2017; Reboredo and Ugolini, 2018; Saeed et al., 2021; Uddin et al., 2019; Wen et al., 2014). Market volatility (Ghosh et al., 2021; Sadorsky,

2022b, 2022a) has proven useful for predicting clean energy stock prices. Following this literature, US stock market volatility (VIX), oil market volatility (OVX), and technology stock market volatility (VXN) are included as features in the analysis.

Economic policy uncertainty (EPU) creates an environment where companies postpone future spending and this affects cash flows and stock prices (Baker et al., 2016; Bekiros et al., 2016, 2016; Pástor and Veronesi, 2012). Sadorsky (2022b) includes economic policy uncertainty as a feature for predicting clean energy stock prices while Wang et al. (2022) use EPU to forecast clean energy stock price volatility. Text based measures of economic policy uncertainty (EPU), equity market uncertainty (EMU), and equity market volatility infectious disease tracker (EMV_IDT) are included in the analysis. Text based measures can provide additional insight to complement market based information (Baker et al., 2016).

Climate change risks are measured using the data created by Faccini et al. (2023). They constructed four measures of climate change risk factors based on textual analysis of around 34,000 articles published by Reuters from January 1, 2000, to December 31, 2018⁶. The Latent Dirichlet Allocation (LDA) textual analysis method was applied to identify topics related to four key climate variables: ‘Natural Disasters,’ ‘Climate Change,’ ‘International Summits,’ and ‘U.S. Climate Policy.’ The first two topics relate to *physical risks*, while the last two refer to *transition risks* due to climate change. In essence, the LDA process decomposes a large collection of documents into a set of topics, and then represents each document as a combination of these topics, with certain probabilities indicating the relevance of each topic to the document. LDA is an unsupervised machine learning method that generates topics. The authors then manually categorize these into 25 distinct topics based on the words that appear most frequently. After

⁶ At the time of writing this paper, the latest version of the climate change risks data ends June 30, 2023.

applying additional criteria, such as the clarity of each topic's interpretation and its relevance to market-wide phenomena, Faccini et al. ultimately identified four key topics. These four topics effectively represent the two dimensions of climate change: physical and transition risks. Finally, the time series of the estimated topic shares for each of the four primary LDA topics are converted into respective risk measures by aggregating the i th topic shares from all articles published on each day. Typically, when news about natural disasters, global warming, or international summits appears in the media, it is viewed as a signal of potential adverse effects on the economy.

Previous studies that predict clean energy stock prices have included technical indicators (Ghosh and Jana, 2024; Sadorsky, 2022b, 2022b, 2022a). Technical indicators are useful for capturing behavioral and momentum factors and have proven important for predicting asset prices (Bustos and Pomares-Quimbaya, 2020; Neely et al., 2014; Sadorsky, 2022b; Wang et al., 2020; Yin et al., 2017; Yin and Yang, 2016). The following technical indicators are included in the analysis; the 50-day and 200-day moving averages (MA50 and MA200), the relative strength indicator (RSI), the moving average cross-over divergence (MACD), stochastic oscillator (slow denoted by StoSLOWD and fast denoted by StoFASTD), advance-decline line (ADX), price rate of change (ROC), on balance volume (OnBalanceVolume), money flow index (MFI), and Williams accumulation and distribution (WAD).

The stock price data are from Yahoo Finance. Interest rates and inflation expectations data are from the St. Louis Federal Reserve Data base (FRED). The EPU, EMV and EMV_IDT data are downloaded from the economic policy uncertainty website (<https://www.policyuncertainty.com/>) (Baker et al., 2016). The climate change risk data are from Renato Faccini's website (<https://sites.google.com/site/econrenatofaccini/home/research>). The

start date for the analysis, June 27, 2008, is determined by the start date of FAN and the ending date June 30, 2023 is determined by the availability of the climate change risks data.

Time series patterns of the clean energy ETFs are displayed in Figure 2. The plot shows a growing trend in the clean energy stocks from 2012 onwards, with solar (TAN) exhibiting strong performance in recent years, despite its high volatility. The wind energy ETF (FAN) displayed a steady, moderate increase over the period, while the nuclear energy ETF (NLR) remained relatively flat, with a slight uptick towards the end. These differing trends among the clean energy ETFs reflect an upsurge in the clean energy market, potentially driven by new policies, technological advances, or shifts in public sentiment towards renewable energy. Notice that FAN and TAN experienced large price appreciation in late 2020 and early 2021. Several factors lead to these price increases including large declines in the costs of producing renewable energy (especially solar), a reduction in electricity demand and more workers working remotely (these sudden shifts favors cheaper renewable electricity generation), greater investor interest (in response to concerns about climate change), and more generous renewable energy policy (part of the COVID relief packages in many countries)⁷.

⁷ <https://www.weforum.org/agenda/2020/06/covid-19-is-a-game-changer-for-renewable-energy/>

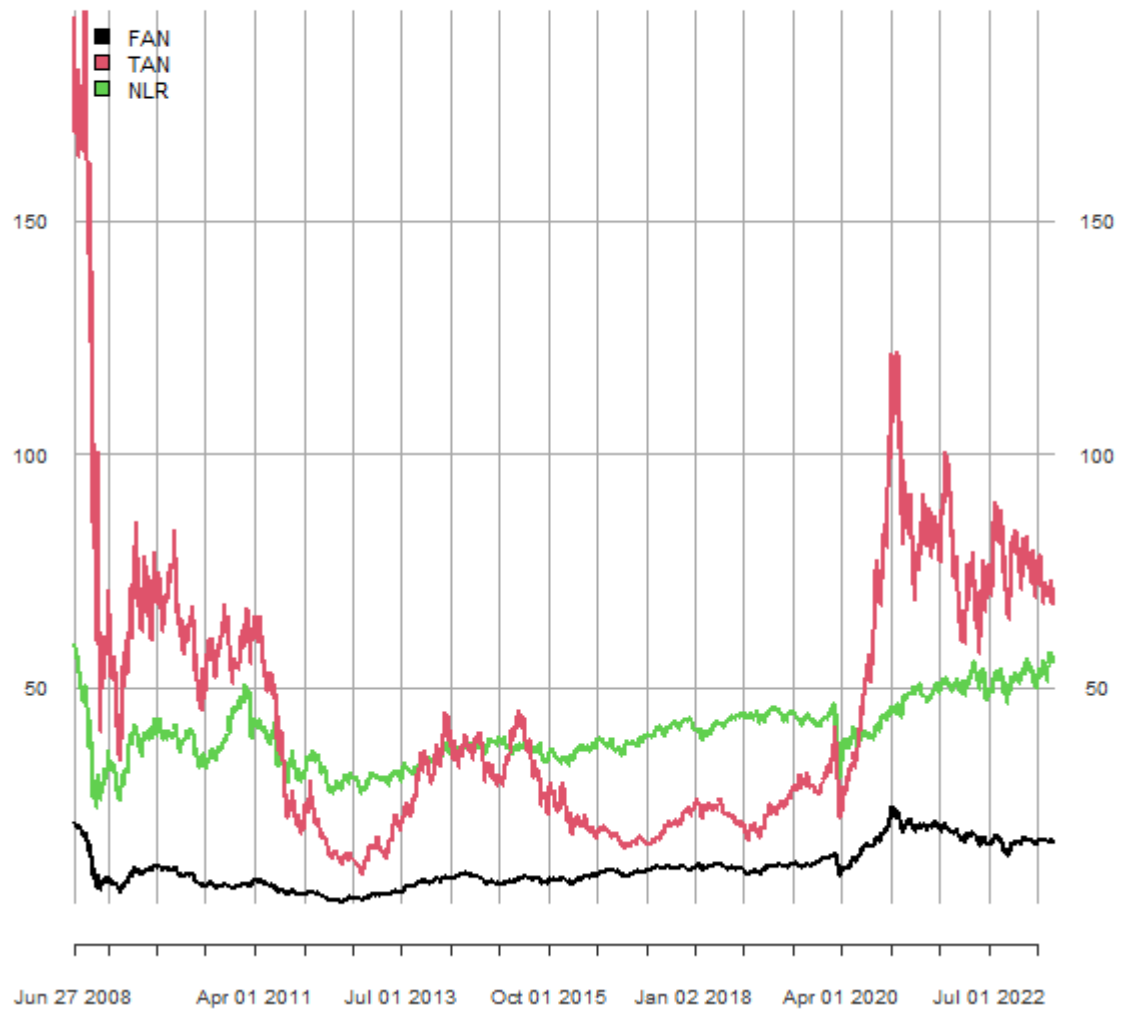


Figure 2. Time series plots of clean energy ETFs.

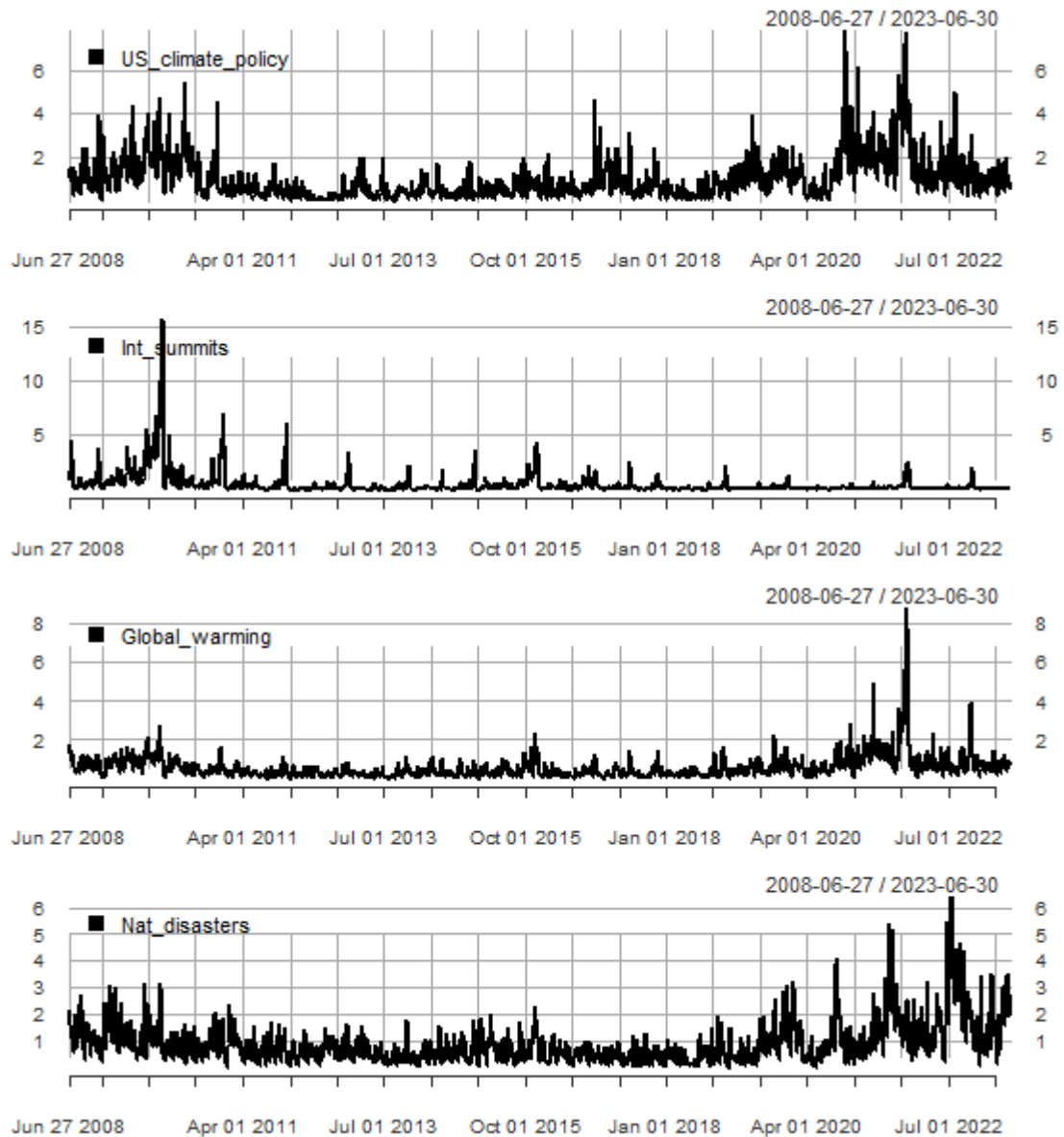


Figure 3. Time series plots of climate change risks (5 day moving average).

Figure 3 presents time series plots of climate change risks, based on a 5-day moving average. Several observations stand out. First, the US climate policy measure displays periodic fluctuations over the 15-year period, with notable clusters of activity around 2016-2017 and in 2020, likely reflecting the influence of US presidential election cycles and related shifts in climate policy. Second, the international summits plot shows intermittent yet significant spikes,

with a sharp peak in 2009 (The Copenhagen Conference) and another in late 2015, corresponding to the Paris Agreement negotiations during COP21. The global warming plot reveals an increasing trend in media or public focus, as shown by the growing frequency of spikes. Lastly, the natural disasters plot shows a rising trend in both the frequency and intensity of peaks, aligning with an increase in the occurrence and severity of such events. Collectively, these time series plots depict the diverse nature of climate change risks and their representation in the media, with global warming and natural disasters receiving progressively more coverage, likely reflecting their escalating real-world impacts. The calculated pairwise correlations between the four climate textual factors are 0.31, indicating that the health of the planet changes very slowly over time. Furthermore, the correlations between climate factors and other economic variables, such as VIX and EPU, also tend to be small (less than 0.30).

Table 1. Summary statistics.

	median	mean	std.dev	coef.var	skewness	kurtosis	normtest.W	normtest.p
FAN	0.052	-0.005	1.739	-323.333	-0.416	12.151	0.882	0.000
TAN	0.019	-0.026	2.883	-109.051	-0.336	6.151	0.934	0.000
NLR	0.000	-0.001	1.480	-1519.568	-0.907	12.955	0.871	0.000
EMU	42.768	63.836	65.773	1.030	3.119	15.000	0.708	0.000
EPU	108.870	125.865	74.527	0.592	2.483	9.194	0.788	0.000
Tenyrbond	2.372	2.413	0.805	0.334	-0.027	-0.501	0.985	0.000
ThreemTbill	0.142	0.738	1.152	1.561	2.075	4.025	0.665	0.000
Fiveyrbeinf	1.772	1.774	0.593	0.334	-1.022	5.288	0.922	0.000
spread	1.708	1.657	1.132	0.683	-0.539	0.267	0.976	0.000
EMV_IDT	0.394	3.520	6.891	1.958	2.928	11.644	0.563	0.000
Fiveyrinfexp	2.194	2.187	0.370	0.169	-0.501	0.866	0.981	0.000
VIX	17.808	20.333	9.338	0.459	2.281	7.160	0.794	0.000
OVX	36.042	39.297	18.126	0.461	3.748	24.873	0.729	0.000
VXN	20.208	22.909	9.147	0.399	1.921	5.427	0.837	0.000
US_climate_policy	0.744	1.040	0.985	0.947	2.247	7.684	0.801	0.000
Int_summits	0.144	0.485	1.055	2.177	5.895	52.728	0.449	0.000
Global_warming	0.414	0.579	0.646	1.116	5.024	44.303	0.639	0.000
Nat_disasters	0.705	0.910	0.785	0.862	2.203	6.932	0.806	0.000

FAN, TAN, and NLR are expressed in continuous compound daily returns. Other variables are five period moving averages. Data are for the period June 27, 2008 to June 30, 2023.

Table 1 presents summary statistics for the variables used in our empirical analysis. The negative average returns for FAN (wind), TAN (solar), and NLR (nuclear power) indicate challenging market conditions in their respective sectors during the analyzed period. The global financial crisis was particularly hard on the clean energy sector as many of the companies involved in funding clean energy experienced high levels of financial stress and bankruptcy. Among the clean energy ETFs, TAN had the highest standard deviation while NLR had the lowest. Each ETF exhibits a distinct volatility profile relative to its performance. Moreover, the combination of negative skewness and high kurtosis in these ETFs points to a risk profile characterized by generally less favorable returns and a higher likelihood of extreme values. This pattern reflects the inherent volatility and risk in the clean energy sector, influenced by factors such as technological advancements, policy changes, and market dynamics.

Among the four climate change risk factors, the relatively high mean values for US climate policy and natural disasters suggest consistent, significant media attention. This is understandable given the US's substantial impact on global emissions and its influential role in international climate politics. In contrast, the lower mean value for international summits reflects their episodic nature, with media attention peaking during significant events. The moderate mean value for global warming aligns with its status as a persistent media topic, covering new scientific findings, impacts on ecosystems and communities, and mitigation and adaptation discussions. Similarly, the high mean value for natural disasters indicates their growing frequency and severity, often associated with climate change. Overall, the high mean values for US climate policy and natural disasters reflect their immediate relevance and impact. In contrast, international summits, while important, tend to attract fluctuating attention, and global warming

maintains a steady presence in the media, indicative of its ongoing and pervasive nature. The standard deviation and coefficient of variation values indicate that, the international summits variable has the most variability while the natural disasters variable has the least. Finally, the extremely high kurtosis values for international summits and global warming imply that, although media coverage for these topics is often moderate or low, there are occasional, very intense spikes in coverage, indicating periods of extremely focused media attention.

Turning to the other variables in the model, indicators like EMU, EPU, EMV_IDT, and OVX show high volatility and a tendency for extreme values, reflecting the dynamic and often unpredictable nature of these variables. In contrast, features such as the 10-year bond yield (Tenyrbond) and five-year inflation expectation (Fiveyrinfexp) exhibit more stable behavior. Finally, the Shapiro-Wilk test strongly rejects the normal distribution for these data.

5. Results

Results are presented first for prediction accuracy and then for variable importance. This section of the paper ends with a portfolio comparison.

5.1. Forecast accuracy

Figure 4 shows the accuracy of the machine learning models in forecasting clean energy stock price direction over a span of 1 to 20 days. The plots indicate that ensemble models like Extra Trees, LightGBM, and Random Forest are superior in forecasting clean energy stock price direction, particularly evident after the 10-day mark. Ensemble models are predominantly good at managing high-dimensional data without falling victim to overfitting, a common challenge in machine learning. Moreover, they can uncover and model complex, non-linear patterns more effectively than simpler, linear models. This implies that clean energy stock price direction

movements are complex, with non-linear patterns better captured by these advanced models. The stabilization or improvement in accuracy with increased forecast horizons indicates that the 10 day to 20 day periods are optimal for model predictions. On the other hand, simpler models such as Lasso and Naive Bayes underperform, highlighting the sophistication required in modeling for investment strategies in clean energy stock prices.

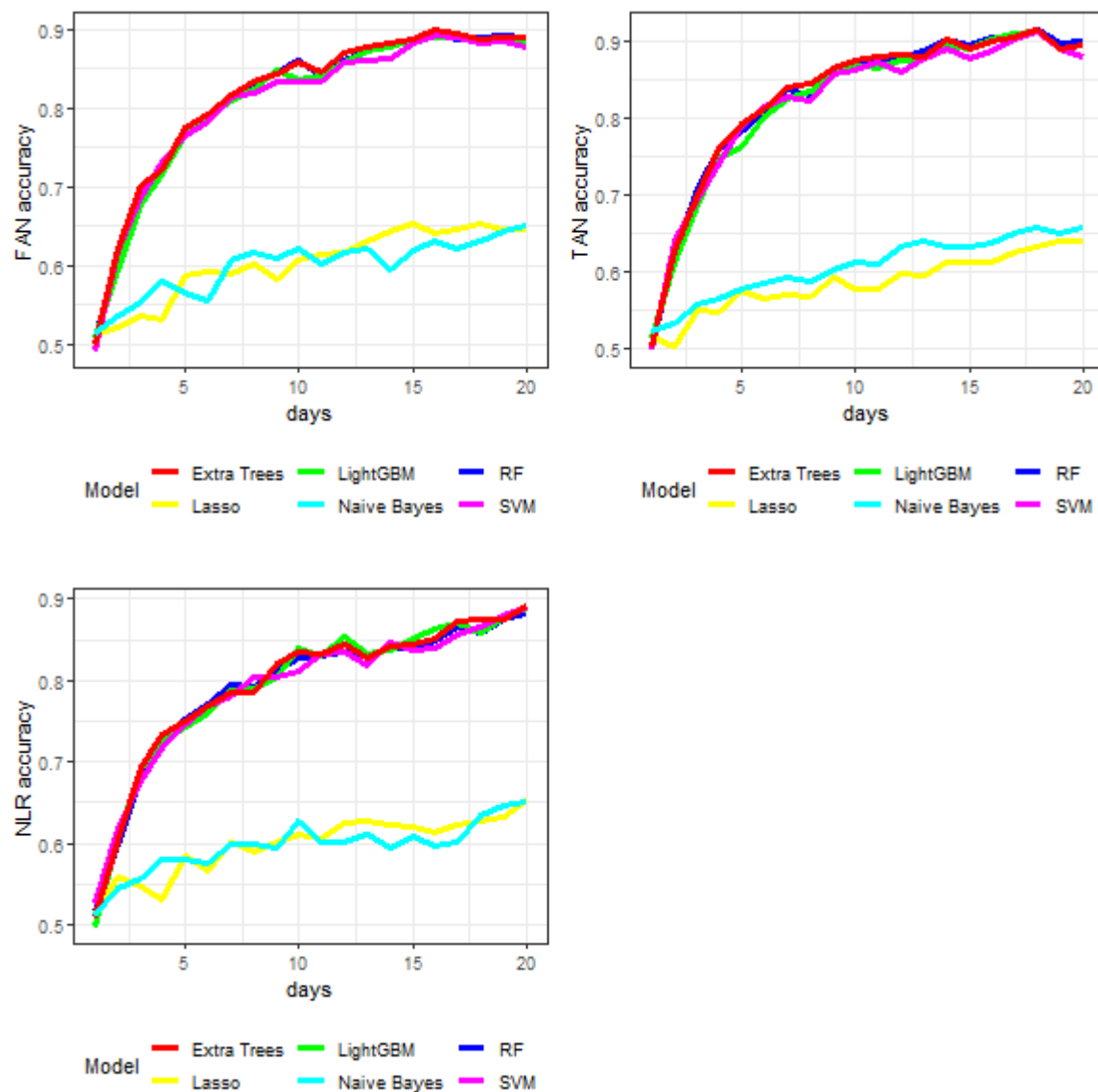


Figure 4. Prediction accuracy. The vertical axis represents accuracy in decimal form and the horizontal axis represents the number of days being forecast.

Figure 5 presents the forecasting accuracy of various machine learning models using Kappa statistics, which measure the degree to which the models outperform random chance. Once again, ensemble models such as Extra Trees, LightGBM, and Random Forest significantly outshine simpler models like Lasso and Naive Bayes particularly in the 10-day to 20-day forecast mark. The consistency in results between high Kappa values and model accuracy enhances the perceived dependability of these models for medium-term market predictions.

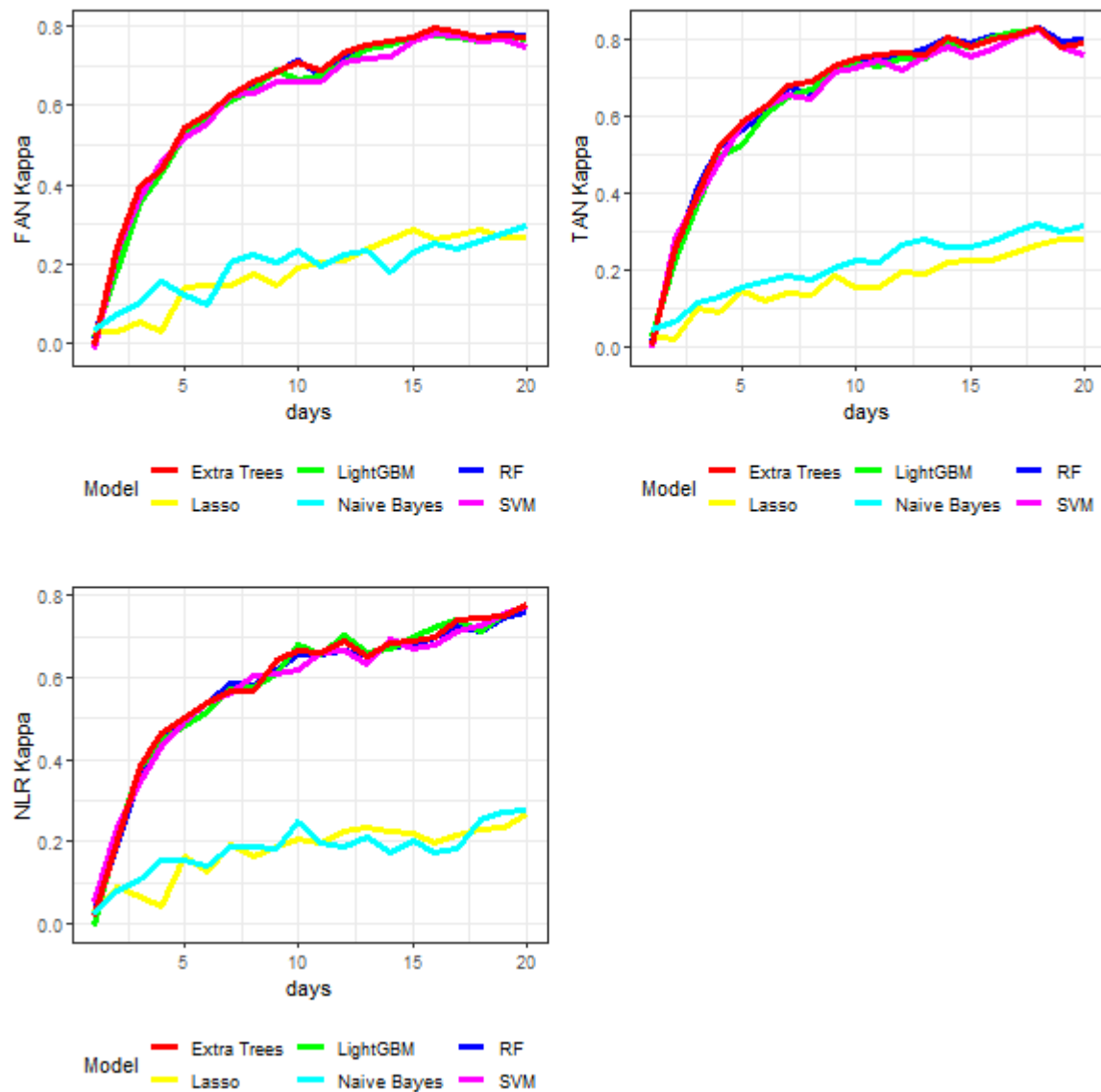


Figure 5. Kappa values. The vertical axis represents accuracy in decimal form and the horizontal axis represents the number of days being forecast.

Figure 6 shows prediction accuracy for different machine learning models based on F1 score, which indicates how well these models balance precision and recall. Not surprisingly, the ensemble models, namely Extra Trees, LightGBM, and Random Forest, show superior F1 scores across all horizons, particularly improving after the 5-day mark and stabilizing beyond 10 days. In contrast, simpler models like Lasso and Naive Bayes demonstrate lower F1 scores, suggesting they might not be as effective in capturing the balance between precision and recall when predicting ETF performances. The overall trend suggests that the complex patterns present in the ETF data are better modeled by ensemble models that can handle the trade-off between recall (capturing all relevant instances) and precision (ensuring relevance of the instances captured).

The appendix contains additional analysis pertaining to statistical significance between differences in predictive accuracy (based on the area under the receiver operator curve) between the models. A Friedman's rank sum test provides statistical evidence that the Extra Trees, LightGBM, RF, and SVM models have similar prediction accuracy.

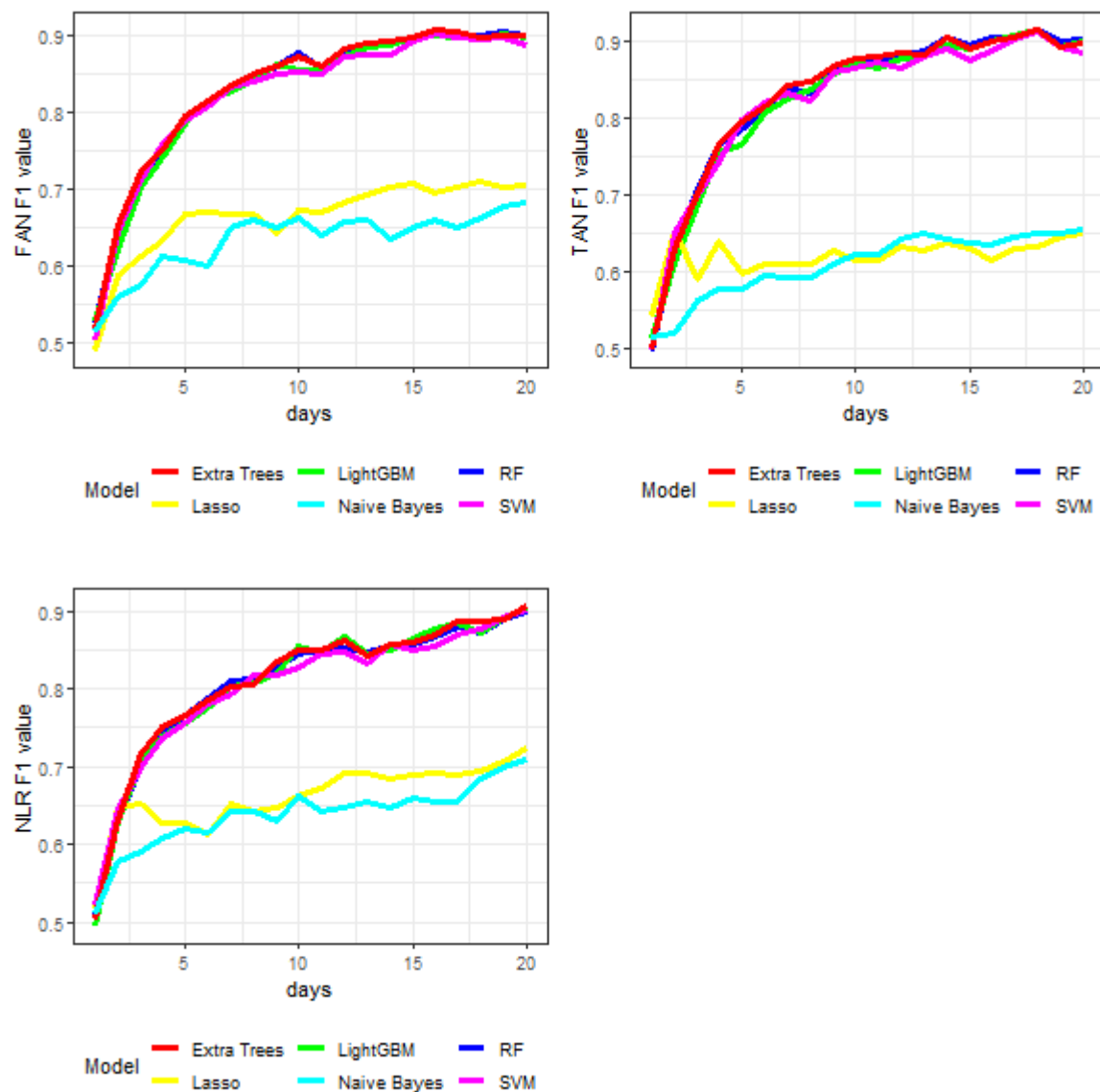


Figure 6. F1 value. The vertical axis represents accuracy in decimal form and the horizontal axis represents the number of days being forecast.

5.2. Feature importance

Feature importance for FAN is calculated using Shapley values (Figure 7a). The numerical values on the horizontal axis are mean absolute Shapley values and correspond to probability values of being in the positive price direction prediction state. Notice that for the 10-day and 20-day forecasts (where forecast accuracy is the highest), Fiveyrinfxp, Fiveyrbeinf,

MA50, OnBalanceVolume, and WAD are ranked in the top six. The two measures of inflation, Fiveyrinexp and Fiveyrbenef, consistently play important roles in predicting price direction for the 5-day, 10-day, and 20-day forecasts. The average of the mean absolute Shapley values for 1 day, 5 day, 10 day, and 20 day forecasts are 0.009, 0.012, 0.014, and 0.016 respectively. These values indicate that, on average, variable importance increases with forecast horizon. At the 20 day forecast horizon, including five year break even inflation as a feature increases the probability of a positive price direction forecast by 3.7% while including WAD as a feature increases the probability of a positive price direction forecast by 3.3%. Any feature ranked higher than MACDSignal is ranked in the top half of importance. OVX is the highest ranked volatility feature for 5, 10, and 20 day forecasts.

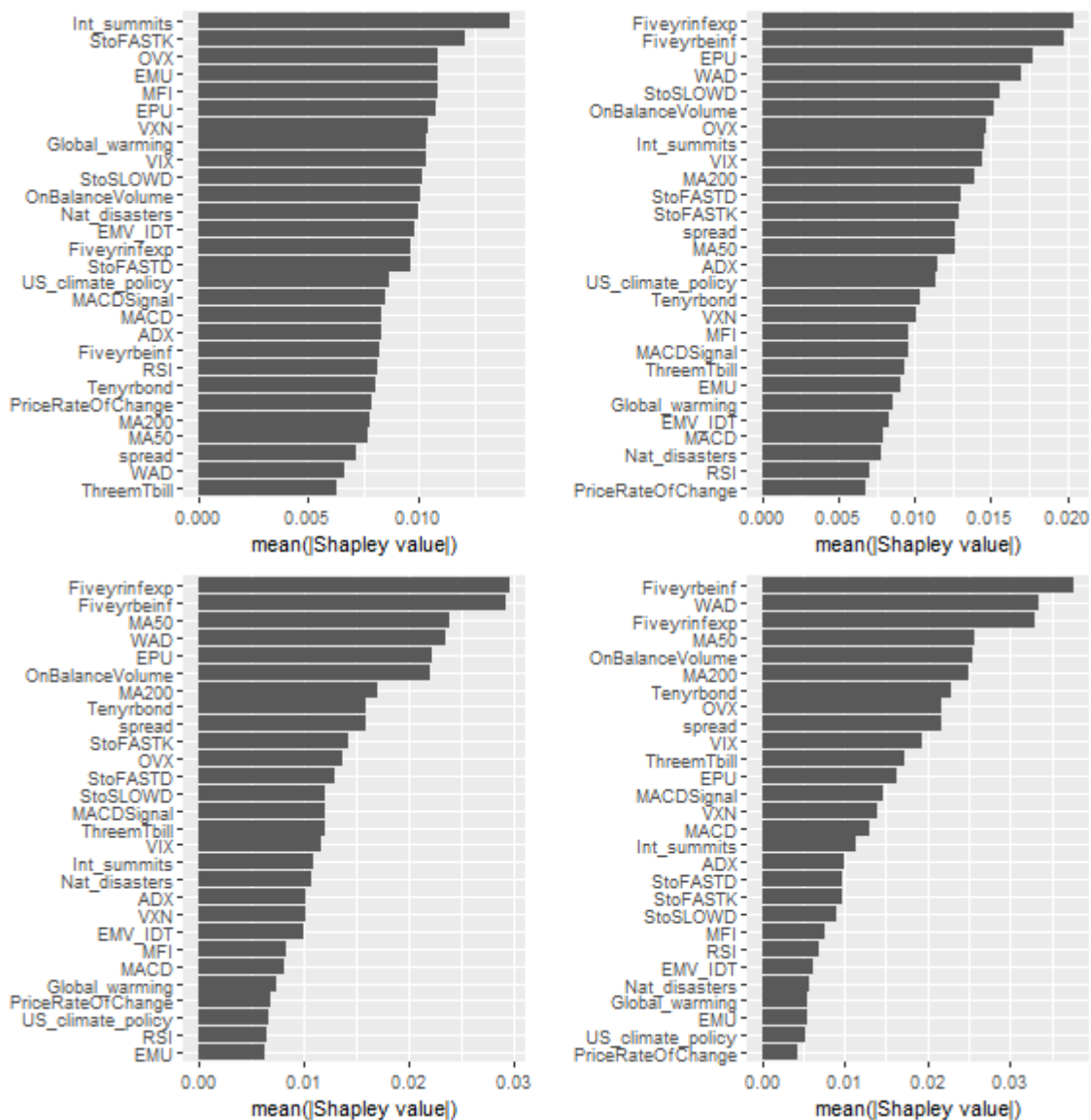


Figure 7a. Shapley values for FAN. Top left (1 day forecast horizon), top right (5 day forecast horizon), bottom left (10 day forecast horizon), and bottom right (20 day forecast horizon). The vertical axis shows the ranking of variables by their importance and the horizontal axis reports values as percentages in decimal form.

As in the case of FAN, Fiveyrinfexp, Fiveyrbeinf, and MA200, are highly ranked features for TAN (Figure 7b). For 20 day forecasts, including Fiveyrbeinf increases the probability of a positive price direction forecast by 4.3% while including Fiveyrinfexp as a feature increases the probability of a positive price direction forecast by 4.1%. The average of the mean absolute

Shapley values for 1 day, 5 day, 10 day, and 20 day forecasts are 0.010, 0.013, 0.014, and 0.016 respectively. For 20 day forecasts, any feature ranked higher than MACDSignal is ranked in the top half of importance. For 5, 10, and 20 day forecasts, VIX is the highest ranked volatility feature.

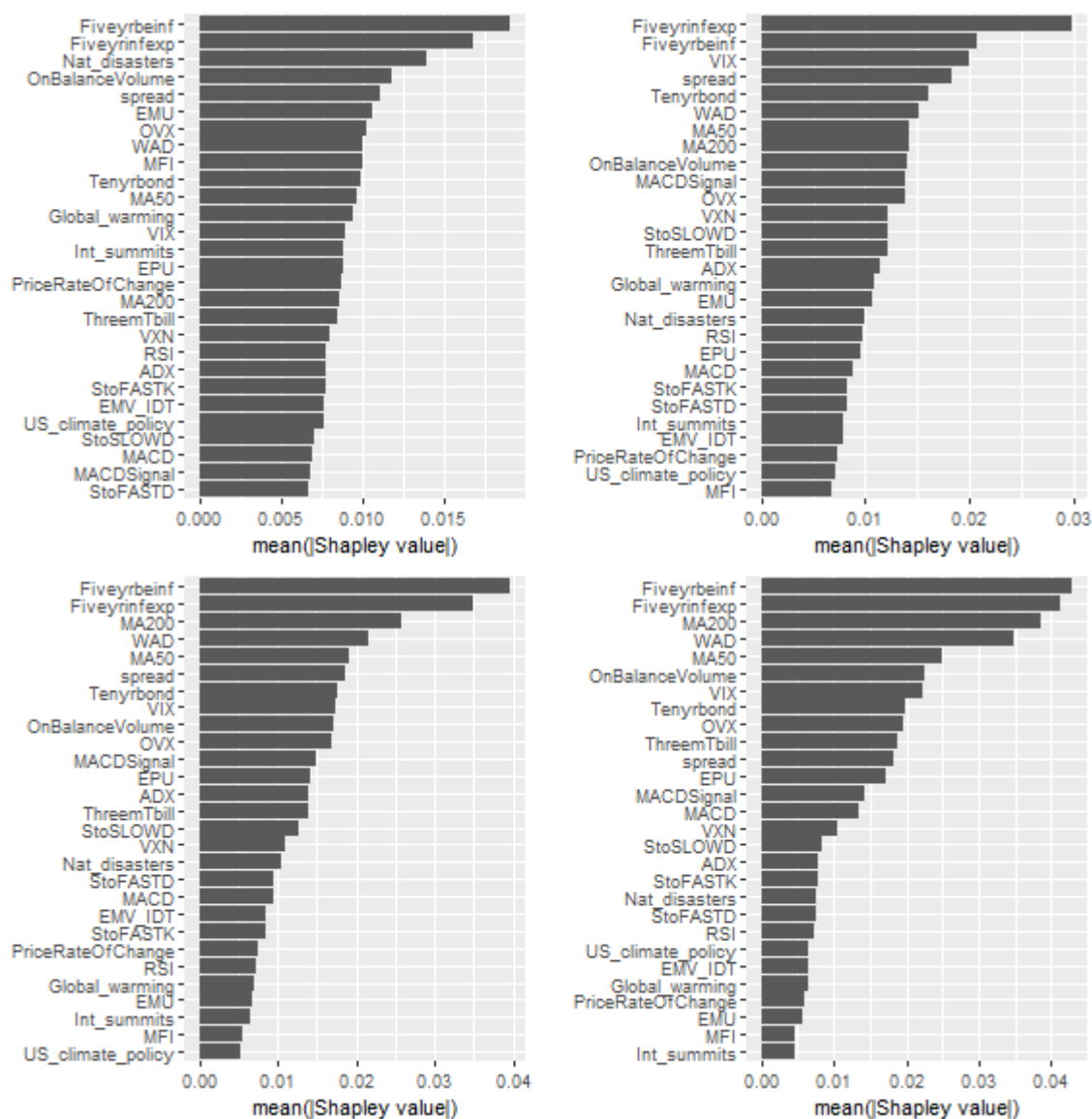


Figure 7b. Shapley values for TAN. Top left (1 day forecast horizon), top right (5 day forecast horizon), bottom left (10 day forecast horizon), and bottom right (20 day forecast horizon). The vertical axis shows the ranking of variables by their importance and the horizontal axis reports values as percentages in decimal form.

Figure 7c shows mean Shapely values for various features across forecast horizons for NLR, a nuclear energy ETF. EPU, VIX, and MACDSignal are highly ranked features for each of the forecast horizons. This consistency suggests that the model heavily considers momentum and economic uncertainty as significant factors in predicting performance. Moreover, the consistency of market volatility indicators in predicting price direction of nuclear ETF is notable. At the 20 day forecast horizon, including MACDSignal as a feature increases the probability of a positive price direction by 3.5%. The average of the mean absolute Shapley values for 1 day, 5 day, 10 day, and 20 day forecasts are 0.009, 0.012, 0.014, and 0.015 respectively. For the 20 day forecast horizon, any feature value ranked higher than the 10 year bond is in the top half of feature importance.

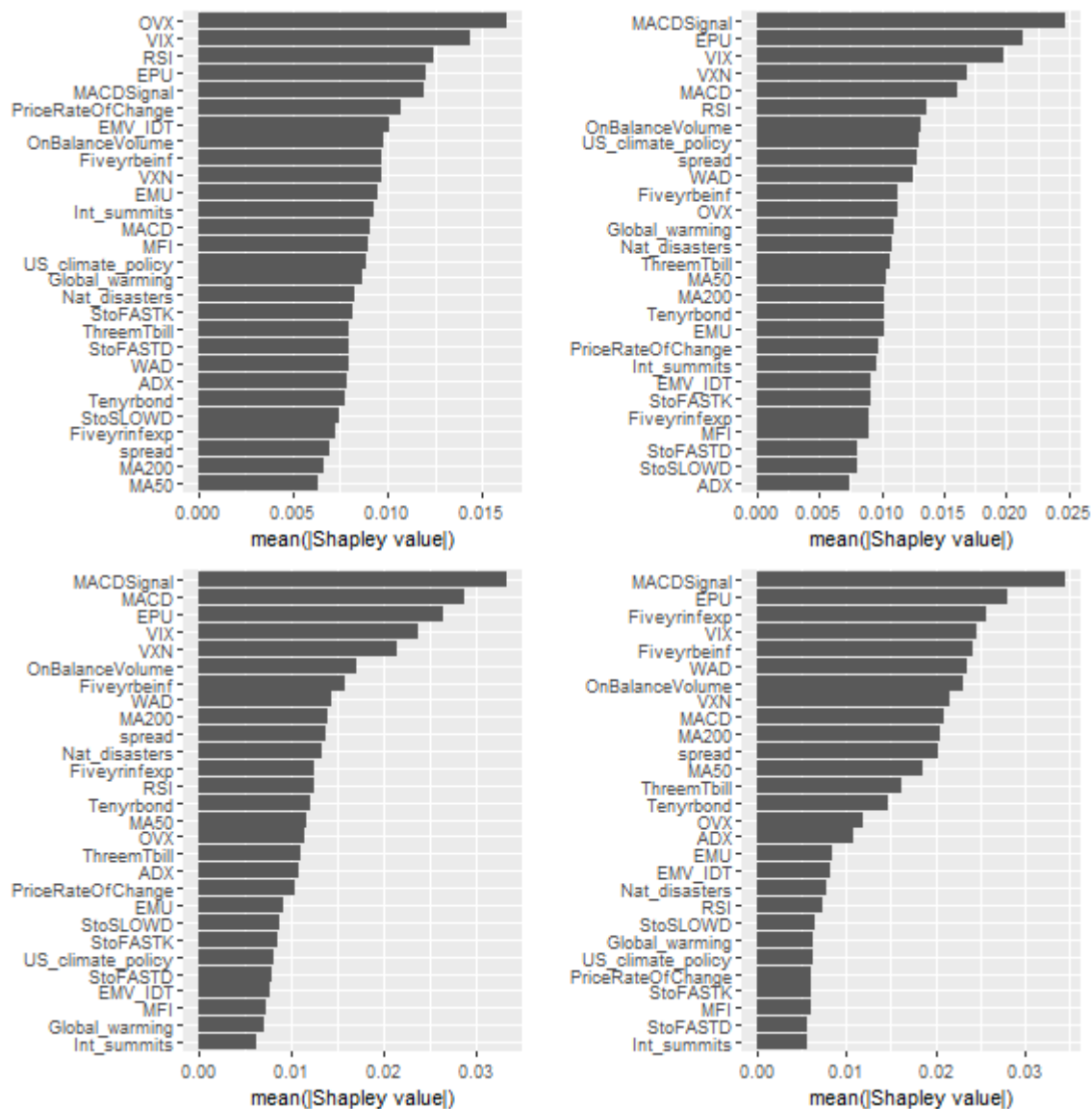


Figure 7c. Shapley values for NLR. Top left (1 day forecast horizon), top right (5 day forecast horizon), bottom left (10 day forecast horizon), and bottom right (20 day forecast horizon). The vertical axis shows the ranking of variables by their importance and the horizontal axis reports values as percentages in decimal form.

In summarizing the results on Shapley values for the five day to twenty day forecast range (reported in Figures 7a, 7b, and 7c), a common pattern is observed, where climate change risk variables are not very important. These findings are consistent with those of others who find equity prices not fully reflecting climate change risks (Bolton et al., 2020; Campiglio et al.,

2022; Eren et al., 2022; IMF, 2020). The appendix provides some additional statistical analysis (based on the area under the receiver operator curve) confirming the relative unimportance of climate change risk features in predicting clean energy stock price direction.

The results also show that FAN and TAN share some commonalities in terms of feature importance. For FAN and TAN inflation features and technical indicators like moving averages, WAD and on-balance volume are highly important for predicting stock price direction in the 10 day to 20 day forecast range, suggesting that inflation concerns and investor sentiment and momentum significantly influence price movements. Moving averages measure price movements while on-balance volume helps gauge investor sentiment through volume trends. These indicators play a key role in assessing price trends and market positioning.

For nuclear energy, economic policy uncertainty and stock market volatility indicators, notably the VIX, underscore the impact of economic and market uncertainty on predicting the nuclear energy ETF for the full range of forecasts. The MACDSignal is the highest ranked technical indicator feature for each forecast horizon. Compared to wind and solar energy, nuclear energy is more sensitive to economic policy uncertainty and stock market volatility. Inflation expectations and other technical indicators (like moving averages) are important features for predicting NLR but not as important as economic policy uncertainty and stock market volatility.

Figure 8 shows SHAP variable importance calculated from a random forest for a 20-day forecast horizon for FAN. The ranking of the variables in Figure 8 is the same as that in Figure 7a (lower right) but Figure 8 includes additional information on the sign of the features. Lower values of Fiveyrbeinf, WAD, Fiveyrinexp, MA50, OnBalanceVolume, MA200, spread, MACDSignal, ADX, MFI, RSI, and EMU increase the probability of being in the positive price prediction state. Higher values of EPU, and VXN increase the probability of being in the positive

price direction state. For the other variables (Tenyrbond, OVX, VIX, ThreemTbill, MACD, Int_summits, StoFASTD, StoFASTK, StoSLOWD, EMV_IDT, Nat_disasters, Global_warming, US_climate_policy, and PriceRateOfChange) the relationship between feature importance and price direction is nonlinear (no clearly identifiable color pattern).

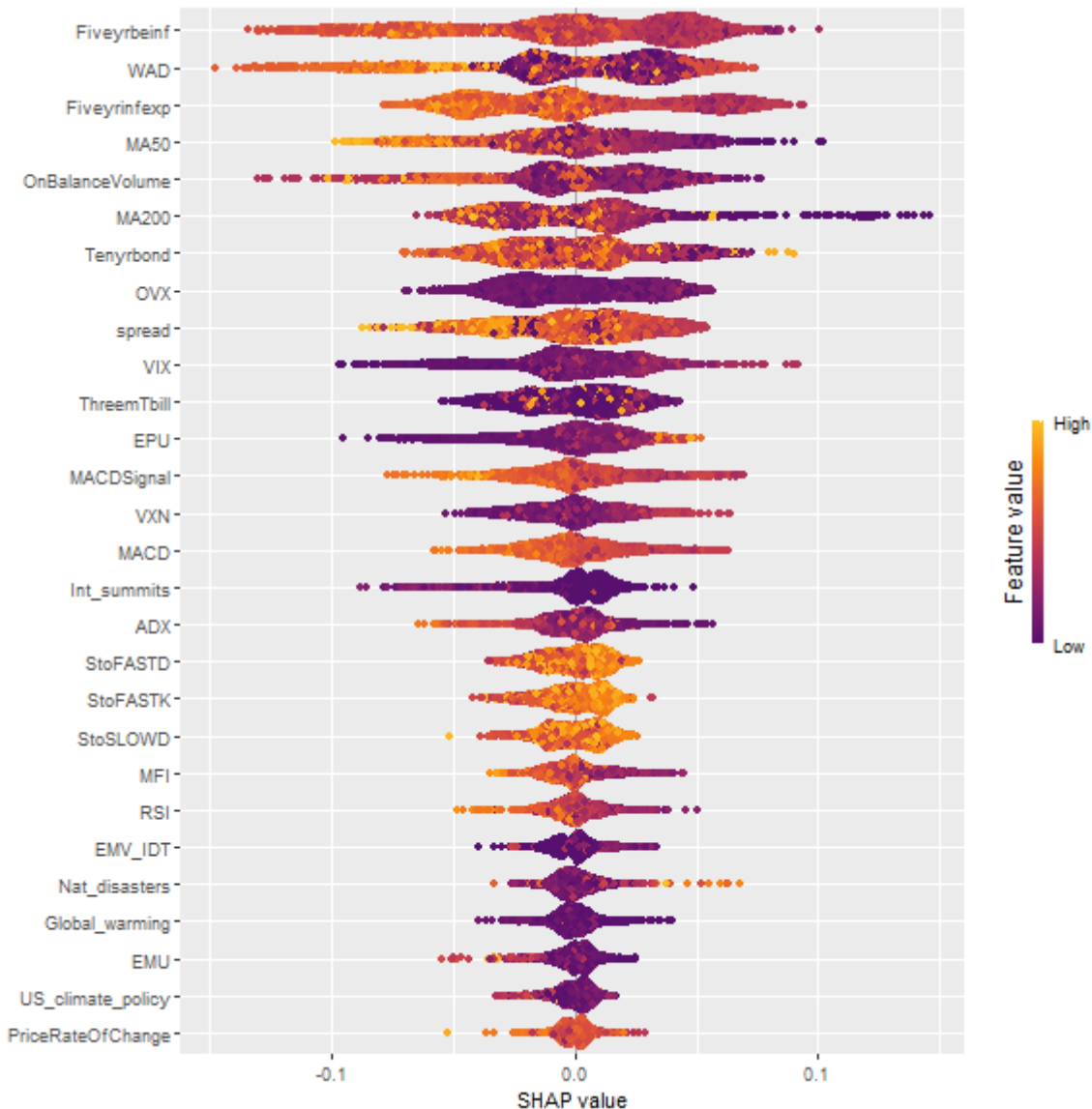


Figure 8. FAN SHAP variable importance (random forest 20 -day forecast horizon). The vertical axis shows the ranking of variables by their importance and the horizontal axis reports values as percentages in decimal form.

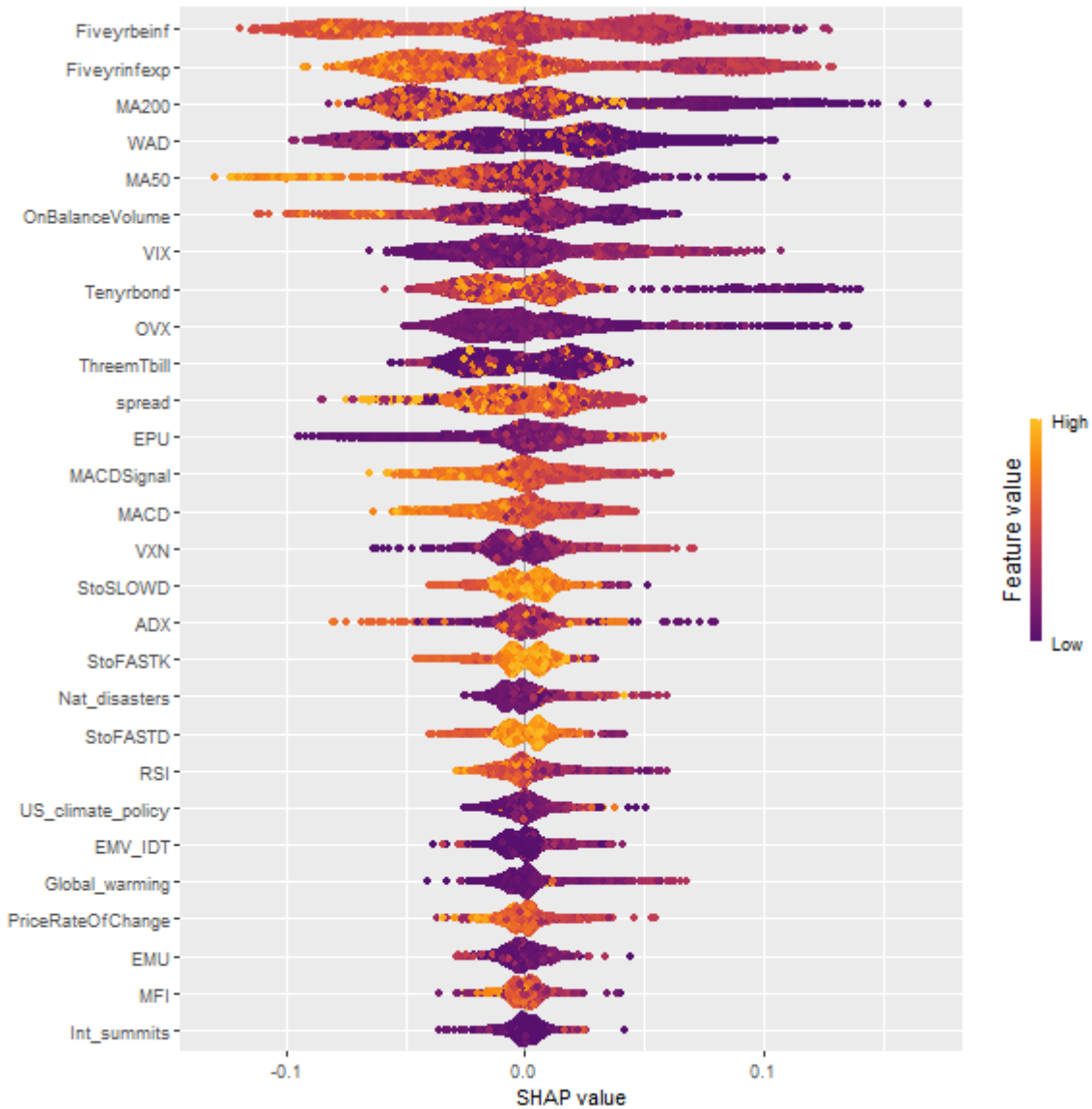


Figure 9. TAN SHAP variable importance (random forest 20-day forecast horizon). The vertical axis shows the ranking of variables by their importance and the horizontal axis reports values as percentages in decimal form.

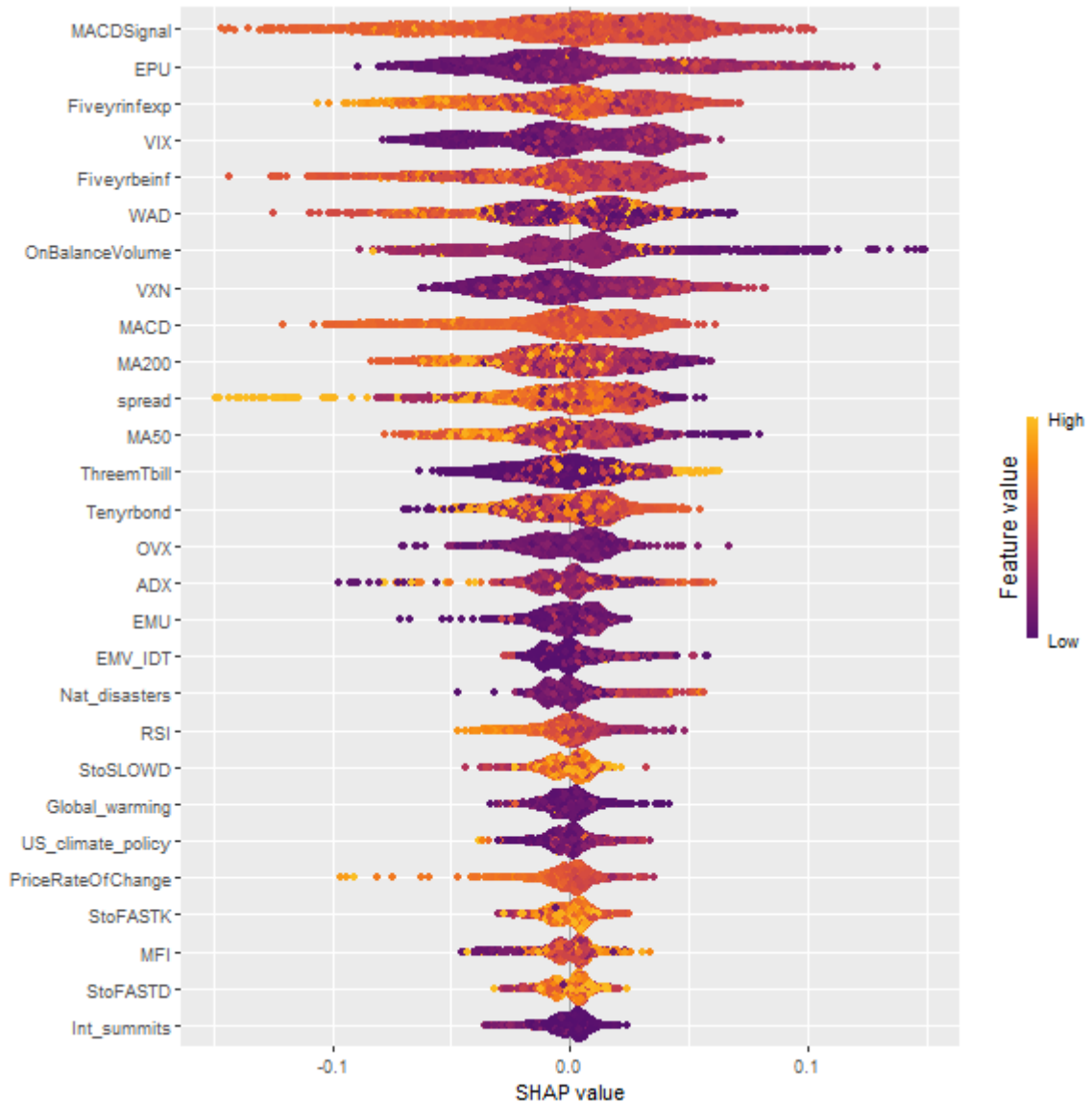


Figure 10. NLR SHAP variable importance (random forest 20-day forecast horizon). The vertical axis shows the ranking of variables by their importance and the horizontal axis reports values as percentages in decimal form.

Figures 8-10 offer a detailed look at how feature value impacts 20-day price direction forecasts for the clean energy ETFs. Looking at Figures 8-10 there are several important observations. First, for all three ETFs, the inflation variables are ranked high in importance and lower values of inflation imply a higher probability of being in the positive price direction state. Second, for each ETF the technical indicators WAD and OnBalanceVolume are ranked high in

importance and lower values increase the probability of being in the positive price direction state. Third, in the case of FAN (Figure 8) and TAN (Figure 9), MA50, and MA200 are highly ranked features and lower values increase the probability of being in the positive price direction state. Fourth, for NLR (Figure 10), MACDSignal, EPU, and VIX are ranked high in importance. Lower values of EPU and VIX increase the probability of being in the positive price prediction state. Nuclear energy shows a pronounced sensitivity to market volatility and policy measures, which is perhaps a reflection of the sector's unique risk profile and the significant regulatory scrutiny it faces. The heightened impact of market uncertainty and risk perception on NLR underscores the critical nature of stability and public policy in the valuation of nuclear energy investments.

5.3. Portfolio comparison

In order to provide some practical insight on the advantages of using machine learning to predict clean energy stock price direction, a comparison was made between a buy and hold (B&H) portfolio and a switching portfolio (Switch_et) that uses the extremely randomized trees model to generate trade signals. The portfolios consist of three equally weighted clean energy ETFs – FAN, TAN, and NLR. For each of these ETFs, 20 day price direction forecasts were computed using Extra Trees. If the price direction prediction was positive an investment is made. If the price direction prediction was not positive no investment is made. This combination of model and forecast horizon was chosen because forecast accuracy is highest in the 10 to 20 day range and Extra Trees has high accuracy and fast computation time. Trading costs are set at 100 basis points which is on the high side given that low cost discount brokers allow ETF trading for very low or in some cases even zero commissions. The portfolios were re-balanced either weekly

or monthly. Portfolios constructed using trading signals based on the Naïve Bayes are also included in the analysis (Switch_nb).

Equity curves for weekly rebalancing show how the value of the portfolios evolve across time (Figure 11). By March 2020 the Switch_et portfolio starts to outperform the B&H and by the end of the sample period the outperformance is dramatic (\$7 for the Switch_et compared to \$2 for the B&H and \$2.5 for the Switch_nb). Using the Extra Trees to generate trading signals greatly improves portfolio performance. Equity curves for monthly rebalancing show a similar pattern (Figure 12).

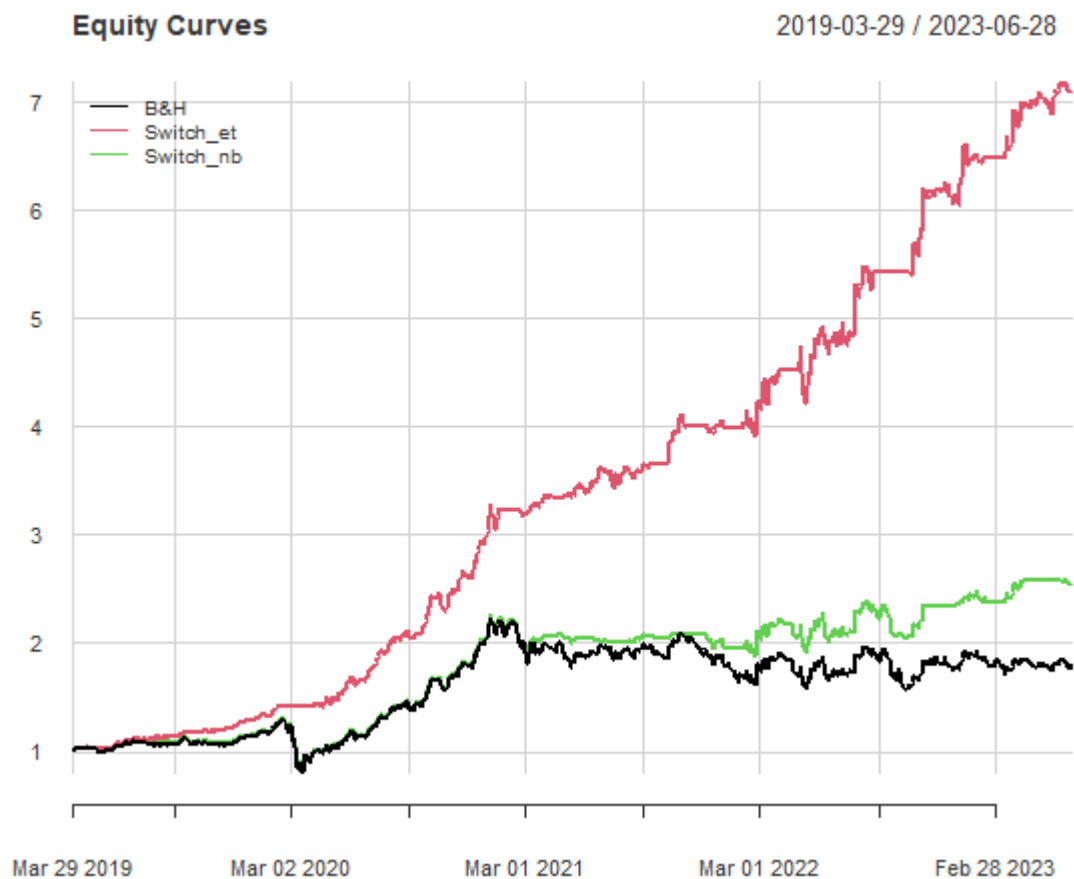


Figure 11. Equity curves for the B&H portfolio, the Switch_nb portfolio and the Switch_et portfolio. The starting values are normalized to \$1 (vertical axis). The equity curves show the cumulative returns across time. Portfolios consist of equally weighted FAN, TAN, and NLR. The Switch_et portfolio uses 20 day predictions from the extremely randomized trees model to generate trade signals. The Switch_nb portfolio uses 20 day predictions from the Naïve Bayes model to generate trade signals. Trading costs are 100 basis points and portfolios are rebalanced weekly.

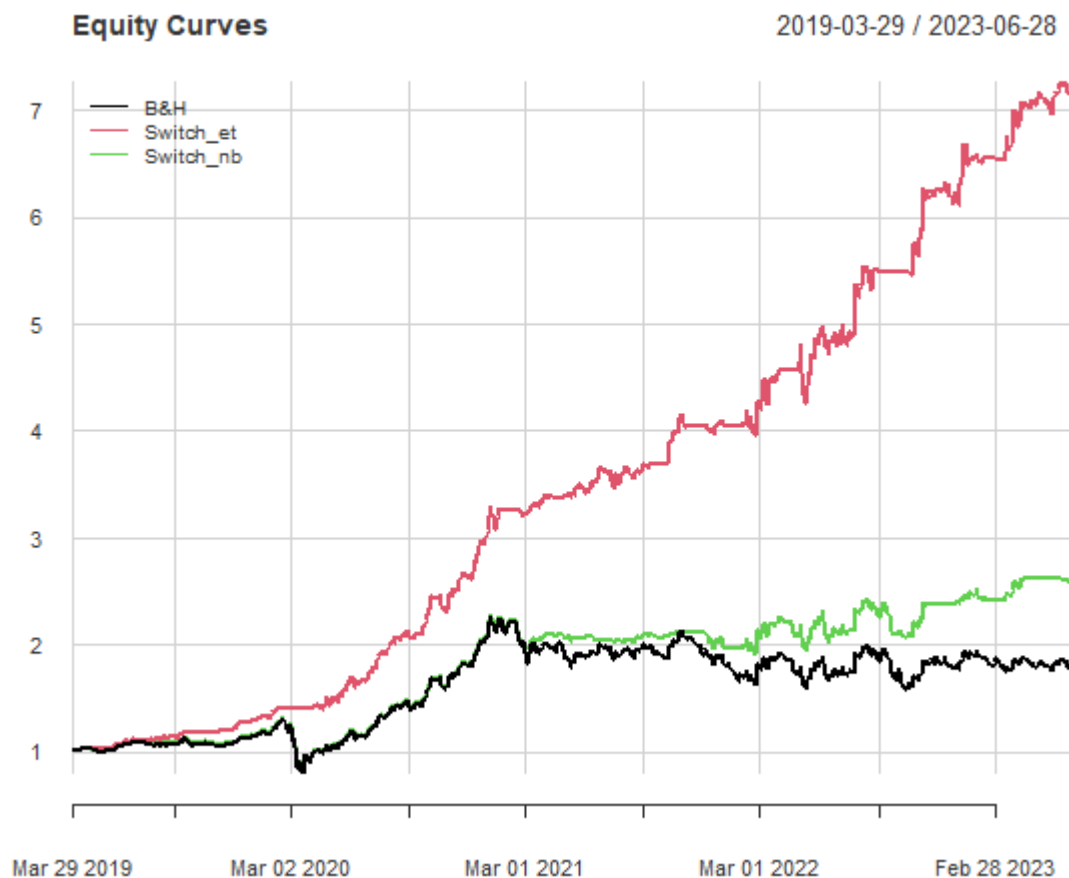


Figure 12. Equity curves for the B&H portfolio, the Switch_nb portfolio and the Switch_et portfolio. The starting values are normalized to \$1 (vertical axis). The equity curves show the cumulative returns across time. Portfolios consist of equally weighted FAN, TAN, and NLR. The Switch_et portfolio uses 20 day predictions from the extremely randomized trees model to generate trade signals. The Switch_nb portfolio uses 20 day predictions from the Naïve Bayes model to generate trade signals. Trading costs are 100 basis points and portfolios are rebalanced monthly.

Portfolio summary statistics show that the average return for the B&H and Switch_et portfolio was 14.7% and 58.8% respectively when rebalancing occurred weekly (Table 2). The Switch_et portfolio generates a higher average return, Sharpe ratio, Omega, and Sortino ratio than the B&H portfolio. The Switch_et portfolio has lower standard deviation, maximum drawdown, and risk measures (value-at-risk (VaR) and expected shortfall (ES)). Notice that the B&H portfolio has a maximum drawdown of 39% while the Switch_et portfolio has a maximum drawdown of 12% - illustrating the ability of the Switch_et portfolio to avoid large losses. Thus, the Switch_et portfolio outperforms the B&H portfolio on a risk adjusted basis. The Switch_nb portfolio outperforms the B&H portfolio but not by as much as the Switch_et portfolio. The summary statistics for portfolios rebalanced monthly also show the Switch_et outperforms the Switch_nb portfolio which outperforms the B&H portfolio.

Table 2. Portfolio summary statistics.

	Weekly			Monthly		
	B&H	Switch_nb	Switch_et	B&H	Switch_nb	Switch_et
Annualized Return	0.147	0.248	0.588	0.150	0.252	0.592
Annualized Std Dev	0.268	0.234	0.160	0.267	0.235	0.161
Annualized Sharpe	0.423	0.899	3.377	0.435	0.918	3.389
Maximum Drawdown	0.387	0.387	0.116	0.388	0.388	0.115
Historical VaR (95%)	-0.023	-0.018	-0.012	-0.023	-0.018	-0.012
Historical ES (95%)	-0.038	-0.035	-0.019	-0.038	-0.035	-0.019
Modified VaR (95%)	-0.027	-0.022	-0.011	-0.027	-0.022	-0.011
Modified ES (95%)	-0.055	-0.051	-0.011	-0.055	-0.051	-0.011
Omega	1.125	1.254	1.860	1.127	1.258	1.860
Sortino Ratio	0.058	0.095	0.345	0.059	0.097	0.345

Portfolios consist of equally weighted FAN, TAN, and NLR. The Switch_et portfolio uses 20 day predictions from the extremely randomized trees model to generate trade signals. The Switch_nb portfolio uses 20 day predictions from the Naïve Bayes model to generate trade signals. Returns are in decimal form. The risk-free rate was set at 3%. Trading costs are 100 basis points and portfolios are rebalanced either weekly or monthly.

6. Discussion

Predicting clean energy stock prices are important for constructing and managing risk of investment portfolios, raising capital for new or existing clean energy companies, and assessing the general outlook for the clean energy sector. The research reported in this paper finds that decision tree-based machine learning models like random forests, extremely randomized trees, and boosting provide higher prediction accuracy over the 3-day to 20-day forecast period than less sophisticated models like Lasso and Naive Bayes. At over 85%, the prediction accuracy of the tree-based ensemble models is particularly impressive over the 10-day to 20-day period. The SVM also has high prediction accuracy. These results support those of Sadorsky (2022a, 2022b, 2021) who also found tree-based ensemble models to have high forecast accuracy for predicting clean energy stock price direction.

Across all ETFs and forecast horizons in the five day to twenty day range, climate change risk measures generally show lower mean Shapley values compared to many other variables in the models. This surprising consistency across different ETFs and forecast horizons suggests that, in the short term, investors might be discounting long-term climate risks in favor of more immediate economic and technical indicators. In other words, the clean energy equity markets (and the financial markets in general) are not pricing in climate change risks. These results align with literature that raises concerns about equity prices not fully reflecting climate change risks (Bolton et al., 2020; Campiglio et al., 2022; Eren et al., 2022; IMF, 2020). Several explanations have been offered for why stock prices may underestimate climate risks. First, the pricing of assets by investors is hampered by uncertainty about the future path of climate change, as well as the energy transition, policy parameters, and adaptation by firms and households (Barnett et al., 2020). Second, instruments to hedge against climate risks are not yet fully developed (Eren et al.,

2022). This is partly because insurance companies are reluctant to sell hedging services due to the uncertain effects of climate change (Bolton et al., 2020). Third, even today, while many scientists view climate change as a prediction about the future, it shapes the perception among people, including investors, that climate change is a distant concern, both in time and space (Oreskes and Stern, 2019). Fourth, investors in clean energy stocks are more oriented to long-term socially responsible investing (El Ghouli and Karoui, 2017). Investors in clean energy stocks are fully aware of the risks that climate change pose and less sensitive to short term measures of climate change risks. Another developing view is that climate change risk is not the same as financial risk⁸. The impact from climate change can be high even when financial risk is low. Under this view, one should not expect climate change risks to be important factors for predicting clean energy stock prices.

In the 10-day to 20-day forecast range, where prediction accuracy is the highest, technical indicators rank high in importance for predicting FAN, TAN, and NLR. This finding is consistent with the findings of (Ghosh and Jana, 2024; Sadorsky, 2022b, 2022b, 2022a). Technical indicators are important because they can capture momentum and behavioral factors.

Inflation expectations rank high in importance for predicting FAN, TAN, and NLR. Investors in clean energy companies are sensitive to inflation expectations because many clean energy stocks are growth stocks that have small current cash flows with the expectation of larger future cash flows. In a low inflation environment, investors are willing to take on additional risk to invest in stocks with the potential to deliver large future gains. In a high inflation environment, investors prefer less risk and switch to companies that are currently profitable and paying dividends. Sadorsky (2022b) also found inflation expectations an important feature for predicting

⁸ <https://robertecclles.com/untangling-climate-risk-financial-risk-and-climate-impact/>

clean energy stock prices. Notice also, that stock market volatility as measured by the VIX ranks in the top third in importance for each of the ETFs studied. This result is similar to Sadorsky (2022b). VIX measures expected stock market fluctuations over the next 30 days. NLR is the only clean energy ETF where EPU ranks high in importance over the 10-day to 20-day range. Nuclear energy's sensitivity to economic policy uncertainty, is perhaps a reflection of the sector's unique risk profile and the significant regulatory scrutiny it faces.

Among non-climate factors and non-technical indicators, macroeconomic and market uncertainty variables have high importance for the movement of clean energy ETF price direction. This is primarily due to the comprehensive information they encompass, such as real economic activity and variables related to financial markets like the stock, commodity, and energy markets, as well as uncertainty measures (Wang et al., 2022). Furthermore, since clean energy supply globally is closely linked to technical innovation (Bai et al., 2020), the global economic condition can substantially affect innovation funding, leading to fluctuations in the clean energy market's supply.

There are several important practical implications from this research. First, forecasting models can safely omit climate change risks as features since they are ranked low in importance. This information is useful for practitioners when choosing features for their forecasting models.

Second, while several tree-based models (random forests, LightGBM, and Extra Trees) rank high on prediction accuracy, our recommendation is to use Extra Trees to predict clean energy stock price direction. These models are preferred over benchmark models like Naïve Bayes and Lasso. The Extra Trees model is particularly impressive because of the high prediction accuracy and relatively fast computational time. This result is of interest to practitioners who want high accuracy predictions without a lot of computational overhead.

Third, a portfolio comparison was used to demonstrate the practical implications of using the predictions made from an Extra Trees model to generate trading signals. This portfolio is referred to as the switching portfolio. More specifically, an investment is made (not made) if the price direction forecasts is positive (non-positive). An equally weighted portfolio of FAN, TAN, and NLR that used trading signals from an Extra Trees prediction model to make investment decision greatly outperformed a buy and hold portfolio in terms of risk adjusted returns. The switching portfolio has the ability to avoid large downside losses and this contributes to higher performance. This result is useful for the design and risk management of clean energy equity portfolios.

7. Conclusions

The clean energy equity sector plays an important role in the transition to a low carbon economy. Accurate forecasts of clean energy stock prices are important for risk managing investments in clean energy stocks and to gauge the outlook for the clean energy sector. This paper uses machine learning to forecast clean energy stock price direction of three clean energy ETFs representing wind, solar, and nuclear energy.

The analysis reported in this paper finds that decision tree-based machine learning models like random forests, extremely randomized trees, and boosting provide higher accuracy over the 3-day to 20-day forecast period than benchmark models like Lasso and Naive Bayes. At over 85%, the accuracy of tree-based ensemble models is particularly impressive for the 10-day to 20-day forecast period. High prediction accuracy is also observed for the SVM. Practitioners who want high clean energy stock price direction prediction accuracy without a lot of computational overhead should use extremely randomized trees. Random forests, which have

comparable accuracy to extremely randomized trees, should be used if a more detailed analysis of feature importance is required.

For ETFs representing solar, wind, and nuclear energy, inflation expectations and technical indicators such as on-balance volume and Williams' accumulation/distribution are important features for forecasting. For wind and solar stocks, moving averages are additional important features while for nuclear energy stocks economic policy uncertainty and the VIX volatility indicator are also important features. Climate change risks are, however, not an important feature for predicting clean energy stock prices. These results are in agreement with literature that raises concerns that equity prices are not fully reflecting climate change risks.

The practical implications of these forecasting results for investors is demonstrated using a portfolio comparison. An equally weighted portfolio of wind, solar, and nuclear energy ETFs that used trading signals from an Extra Trees prediction model greatly outperformed a buy and hold portfolio in terms of risk adjusted returns.

There are several limitations to this research. First, due to data availability the sample period was relatively short and included the COVID-19 period, which may have affected the results due to reduced economic activity. Second, the sample period included the 2020 US election, the outcome of which resulted in a climate friendly president (Joe Biden) who proposed a \$2 trillion climate fund which boosted interest in clean energy stocks. Future presidents may not be as supportive of clean energy. Third, the analysis focused on three important sub-sectors of clean energy (wind, solar, and nuclear) but other sub-sectors like biomass and geothermal were not studied.

Future research could conduct additional analysis in the aftermath of the COVID-19 period as more data becomes available. Future research could also expand the analysis to include other clean energy sub-sectors.

Appendix

The forecast accuracy, kappa, and F1 plots provide practical information on how forecast accuracy compares between different models. While these measures are widely used to compare forecast accuracy from classification machine learning models, these accuracy measures don't, however, provide a statistical analysis of the difference in prediction accuracy between models. To address this issue, receiver operator curves (ROCs) were estimated and the area under the curve (AUC) computed. Pairwise tests for the difference in AUC were computed where the null hypothesis is no difference in AUC and the alternative hypothesis is a difference (Rainio et al., 2024). Probability values were estimated using 2000 bootstrap replications. Test results are provided in Tables A1 – A3.

At the one day forecast horizon, there is no statistically significance difference in AUC values for any of the pairs of models. These results are in accordance with the accuracy plots where all the models have similar accuracy for each of the ETFs studied. In the 10-day to 20-day range a clear pattern emerges where the difference in AUC values between RF, ET, LGBM, and SVM and either Lasso or NB is positive and statistically significant. This provides statistically significance evidence for using a non-benchmark model (RF, ET, LGBM, SVM) over a benchmark model (Lasso, NB).

For 20-day price prediction forecasts for FAN, RF-LGBM, ET-LGBM, and Lasso-NB are pairs where the difference between AUC values is statistically insignificant (Table A1). For

the case of TAN, there is no statistically significance difference in AUC values between RF-LGBM (Table A2). In the case of NLR, there is no statistically significance difference in AUC values between the following pairs: RF-SVM, RF-LGBM, Lasso-NB, SVM-ET, and SVM-LGBM (Table A3).

Table A1. FAN – Difference in AUC pairwise model comparison.

FAN	1 day		5 day		10 day		20 day	
	D	p-value	D	p-value	D	p-value	D	p-value
RF-Lasso	-0.101	0.920	14.073	0.000	18.102	0.000	16.785	0.000
RF-SVM	-0.207	0.836	1.152	0.249	2.842	0.004	4.081	0.000
RF-ET	-0.264	0.791	7.204	0.000	5.456	0.000	3.291	0.001
RF_NB	-0.896	0.370	13.832	0.000	16.342	0.000	16.050	0.000
RF-LGBM	-1.134	0.257	0.003	0.998	1.851	0.064	1.346	0.178
Lasso-SVM	-0.160	0.873	-11.702	0.000	-15.553	0.000	-14.830	0.000
Lasso-ET	-0.242	0.809	-12.537	0.000	-17.694	0.000	-16.771	0.000
Lasso_NB	-0.891	0.373	-0.815	0.415	-2.237	0.025	-1.374	0.170
Lasso-LGBM	-0.363	0.716	-13.120	0.000	-16.499	0.000	-16.118	0.000
SVM-ET	-0.308	0.758	1.688	0.091	-0.406	0.685	-2.520	0.012
SVM_NB	-0.400	0.689	11.941	0.000	13.964	0.000	13.956	0.000
SVM-LGBM	-0.110	0.912	-1.182	0.237	-1.515	0.130	-3.564	0.000
ET_NB	-0.303	0.762	12.198	0.000	15.817	0.000	16.025	0.000
ET-LGBM	-0.027	0.978	-3.765	0.000	-1.246	0.213	-0.984	0.325
NB-LGBM	0.368	0.713	-13.149	0.000	-14.849	0.000	-15.349	0.000

D statistic (difference in AUC values) p-values computed using 2000 bootstrap replications.

Table A2. TAN – Difference in AUC pairwise model comparison.

TAN	1 day		5 day		10 day		20 day	
	D	p-value	D	p-value	D	p-value	D	p-value
RF-Lasso	-0.453	0.651	13.678	0.000	19.417	0.000	17.375	0.000
RF-SVM	-0.509	0.611	1.701	0.089	2.717	0.007	4.363	0.000
RF-ET	-0.156	0.876	2.766	0.006	5.075	0.000	3.569	0.000
RF_NB	-0.504	0.615	13.190	0.000	17.438	0.000	15.975	0.000
RF-LGBM	0.014	0.989	2.103	0.035	1.371	0.170	-1.940	0.052
Lasso-SVM	0.218	0.827	-11.718	0.000	-17.573	0.000	-15.027	0.000
Lasso-ET	0.372	0.710	-13.366	0.000	-18.843	0.000	-17.059	0.000

Lasso_NB	-0.126	0.899	-1.727	0.084	-2.528	0.011	-2.333	0.020
Lasso-LGBM	0.627	0.531	-12.338	0.000	-18.442	0.000	-17.354	0.000
SVM-ET	0.027	0.978	-0.347	0.728	-0.462	0.644	-3.183	0.001
SVM_NB	-0.269	0.788	11.319	0.000	15.288	0.000	12.888	0.000
SVM-LGBM	0.218	0.828	-0.136	0.892	-1.840	0.066	-4.983	0.000
ET_NB	-0.459	0.646	12.697	0.000	16.645	0.000	15.773	0.000
ET-LGBM	0.517	0.605	0.251	0.802	-1.858	0.063	-3.479	0.001
NB-LGBM	0.707	0.480	-11.710	0.000	-16.479	0.000	-15.996	0.000

D statistic (difference in AUC values) p-values computed using 2000 bootstrap replications.

Table A3. NLR – Difference in AUC pairwise model comparison.

NLR	1 day		5 day		10 day		20 day	
	D	p-value	D	p-value	D	p-value	D	p-value
RF-Lasso	-0.747	0.455	12.258	0.000	16.589	0.000	15.644	0.000
RF-SVM	-1.625	0.104	0.492	0.623	3.848	0.000	0.848	0.397
RF-ET	-0.008	0.994	4.089	0.000	3.162	0.002	2.981	0.003
RF_NB	0.051	0.959	11.665	0.000	14.394	0.000	15.729	0.000
RF-LGBM	0.527	0.598	1.586	0.113	1.885	0.059	-0.553	0.581
Lasso-SVM	-0.109	0.913	-11.328	0.000	-13.171	0.000	-14.427	0.000
Lasso-ET	0.759	0.448	-11.604	0.000	-16.427	0.000	-15.284	0.000
Lasso_NB	0.955	0.339	-0.788	0.431	-1.449	0.147	0.008	0.994
Lasso-LGBM	1.328	0.184	-11.375	0.000	-15.430	0.000	-15.506	0.000
SVM-ET	1.980	0.048	1.326	0.185	-2.467	0.014	0.392	0.695
SVM_NB	1.058	0.290	10.891	0.000	11.346	0.000	14.818	0.000
SVM-LGBM	1.167	0.243	0.691	0.490	-2.997	0.003	-1.293	0.196
ET_NB	0.055	0.956	10.932	0.000	14.048	0.000	15.397	0.000
ET-LGBM	0.499	0.618	-0.795	0.427	-0.145	0.885	-2.149	0.032
NB-LGBM	0.606	0.545	-10.663	0.000	-13.350	0.000	-15.513	0.000

D statistic (difference in AUC values) p-values computed using 2000 bootstrap replications.

In terms of prediction accuracy, there appears to be two groups of models – the benchmark models (Lasso, NB) (low prediction accuracy) and the non-benchmark models (RF, ET, LGBM, SVM) (high prediction accuracy). A Friedman’s rank sum test (Rainio et al., 2024) testing the equality of the AUCs from the non-benchmark models (RF, ET, LGBM, SVM) turns up p-values of 0.127, 0.440, and 0.441 for FAN, TAN, and NLR respectively indicating that according to this test there is no statistically significant difference in AUC values between these

models. The results from a Friedman's rank sum test provides statistical evidence that these models have similar prediction accuracy.

As a final look at the importance of the climate change risks features, forecasts were computed from a random forests model that only used the climate change risks as features. The forecast accuracy of the restricted model was compared with the forecast accuracy of the unrestricted model that uses all of the features. For forecast horizons of 5, 10, and 20 days, the prediction accuracy of the restricted model is lower than that of the unrestricted model (Table A4). The forecast accuracy of the restricted model never reaches higher than 66% providing evidence on the low importance of the climate change risks variables.

Table A4. Comparing the forecast accuracy between the restricted and unrestricted random forests model.

		1	5	10	20
FAN	Unrestricted	0.51	0.77	0.86	0.89
	Restrict	0.50	0.59	0.62	0.65
TAN	Unrestricted	0.50	0.78	0.87	0.90
	Restrict	0.47	0.58	0.62	0.65
NLR	Unrestricted	0.51	0.75	0.83	0.88
	Restrict	0.51	0.58	0.62	0.66

Random forest forecast accuracy is reported for 1, 5, 10, and 20 day forecasts. The unrestricted model uses all of the features. The restricted model only uses the climate change risk features.

Pairwise tests for the difference in AUC values between the unrestricted random forests model and the restricted random forests model indicates no statistically significance difference for one day forecasts (Table A5). For forecasts of 5, 10 and 20 days, the difference in AUC values is statistically significant and the positive sign of the difference shows the unrestricted model performs better than the restricted model providing statistical evidence that the unrestricted model predicts more accurately than the restricted model.

Table A5. Difference in AUC pairwise model comparison (unrestricted random forest – restricted random forest).

	1	5	10	20
FAN				
D	-0.333	11.770	15.713	17.217
p-value	0.739	<0.001	<0.001	<0.001
TAN				
D	-1.806	13.608	15.755	16.059
p-value	0.071	<0.001	<0.001	<0.001
NLR				
D	-0.099	12.494	14.525	16.761
p-value	0.921	<0.001	<0.001	<0.001

Random forest forecast accuracy is reported for 1, 5, 10, and 20 day forecasts. The unrestricted model uses all of the features. The restricted model only uses the climate change risk features. The D statistic (difference in AUC values) p-values computed using 2000 bootstrap replications.

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